

Supporting Information

Forecasting Sectoral Crash Risk and Contagion: Integrating Dynamic Volatility, Extreme-Value Modelling and Graph Deep Learning

This Supporting Information reports the diagnostics, implementation details, graph placebos, component ablations, crash-definition robustness, and sector-by-horizon evidence supporting the main manuscript. The primary analysis uses the 2.5% GJG-Runkle generalised autoregressive conditional heteroskedasticity (GJR-GARCH)-filtered extreme-value theory (EVT) crash definition across six pre-specified sectors and chronological Train/Validation/Test partitions. Forecast evaluation covers 1-, 5-, 10-, and 22-trading-day probability horizons. The extended results preserve the main article's focus while documenting the empirical checks needed for transparency and reproducibility.

The document contains SI A–G, Tables S1–S14, and Figures S1–S4. Unless a caption states otherwise, all forecast comparisons use the locked Test period, and lower Brier Score and Log Score indicate better probability forecasts. Negative Dynamic-minus-benchmark Brier differences favour the Dynamic Hawkes Graph model, whereas negative Full-minus-ablation Brier differences favour the Full specification.

SI A. Data construction and sector eligibility

The eligibility screen combines minimum constituent depth with daily coverage requirements to define sectors suitable for the primary analysis. Table S1 reports the full sector screen and summarises the crash profile of the six sectors that satisfy the pre-specified eligibility rule.

Table S1. Sector eligibility and primary-sample composition

Panel A: All sectors

Sector	Stocks	Eligible coverage (%)	Primary	Eligibility decision
Agriculture	6	67.04	No	Eligible sector-day coverage below 90%
Automobiles and Accessories	1	0.00	No	Fewer than 3 constituent stocks
Banking	10	99.88	Yes	Primary
Commercial and services	11	98.87	Yes	Primary
Construction and Allied	2	27.37	No	Fewer than 3 constituent stocks
Energy and Petroleum	4	99.00	Yes	Primary
Insurance	6	99.92	Yes	Primary
Investment	3	99.60	Yes	Primary
Investment Services	1	0.00	No	Fewer than 3 constituent stocks
Manufacturing and Allied	7	92.81	Yes	Primary
Telecommunication and Technology	1	0.00	No	Fewer than 3 constituent stocks

Note. A primary sector requires at least three constituent stocks and at least 90% eligible-day coverage. Sector-day eligibility additionally requires at least two valid returns and at least two-thirds constituent-return coverage.

Panel B: Primary-sector crash profile

Sector	Observed labels	Crash events	Crash rate (%)	Median stocks with return	Median constituent coverage (%)	Mean EVT shape	SD EVT shape
Banking	1985	43	2.17	10	100	0.233	0.042
Commercial and services	1960	41	2.09	9	100	0.033	0.164
Energy and Petroleum	1963	44	2.24	4	100	0.156	0.039
Insurance	1986	55	2.77	6	100	0.126	0.055
Investment	1978	56	2.83	3	100	0.019	0.064
Manufacturing and Allied	1809	41	2.27	4	100	0.021	0.075

Note: Crash events use the primary 2.5% EVT definition. EVT shape summaries use rolling parameter estimates.

Applying the eligibility rules retains six sectors with consistently high eligible-day coverage, while the excluded sectors fail either the constituent-count or coverage requirement. Crash rates range from 2.09% to 2.83% across the retained sectors, showing that crash incidence is broadly distributed rather than concentrated in one sector. Mean EVT shape estimates differ across sectors, while their dispersion also varies, indicating heterogeneous tail behaviour across the retained sectors. Table S2 reports the corresponding minimum, maximum, scale, threshold, and exceedance-rate diagnostics.

SI B. Dynamic volatility and EVT diagnostics

The extreme-value theory (EVT) diagnostics assess whether rolling tail estimates remain internally coherent across the pre-specified crash probabilities. Table S2 summarises 535 rolling refits based on the generalised Pareto distribution (GPD) and reports the resulting 1%, 2.5%, and 5% crash definitions.

Table S2. EVT parameter summaries and threshold robustness

Panel A: Rolling GPD estimates

Sector	Refits	Shape means	Shape SD	Shape min	Shape max	Scale mean	Threshold mean	Mean exceedance rate (%)
Banking	91	0.232	0.043	0.165	0.374	0.575	1.108	9.98
Commercial and services	90	0.034	0.164	-0.361	0.190	0.592	1.163	9.98
Energy and Petroleum	90	0.156	0.039	0.108	0.265	0.562	1.116	9.98
Insurance	91	0.125	0.055	-0.096	0.210	0.627	1.047	9.98
Investment	90	0.019	0.065	-0.122	0.148	0.825	0.966	9.98
Manufacturing and Allied	83	0.021	0.075	-0.112	0.142	0.672	0.991	9.98

Note. Threshold means refer to the 90th-percentile adverse standardised-innovation threshold used for the peaks-over-threshold fit.

Panel B: Crash-rate and return-threshold summary

Tail probability	Observed labels	Crash events	Event rate (%)	Mean crash threshold (%)	Median crash threshold (%)
1%	11681	116	0.993	-4.641	-4.142
2.5%	11681	280	2.397	-3.477	-3.083
5%	11681	582	4.982	-2.654	-2.346

Note. Return thresholds are expressed in percentage-return units; more negative values represent more severe crash thresholds.

Across the six primary sectors, the rolling Generalised Pareto Distribution (GPD) estimates vary over time and across sectors, while the mean exceedance rate remains close to 10% under the 90th-percentile threshold. The pooled crash rates of 0.993%, 2.397%, and 4.982% closely track the corresponding nominal tail probabilities. Return thresholds also follow the expected ordering, becoming less negative as the tail probability increases from 1% to 5%. These diagnostics support retaining the pre-specified 2.5% definition for the primary analysis and the 1% and 5% definitions for robustness.

The following Table S3 evaluates the volatility filter and the integrity checks completed before forecasting.

Table S3. GARCH/EVT integrity and diagnostic checks

Panel A: sector-level diagnostics

Sector	Valid GARCH rows	Standardised innovation means	Standardised innovation SD	EVT refits	EVT convergence (%)	Boundary fits (%)
Banking	2485	0.022	0.988	91	100.0	0.0
Commercial and services	2460	0.033	1.017	90	100.0	0.0
Energy and Petroleum	2463	0.035	0.980	90	100.0	0.0
Insurance	2486	0.013	0.983	91	100.0	0.0
Investment	2478	-0.006	0.958	90	100.0	0.0
Manufacturing and Allied	2309	0.049	1.000	83	100.0	0.0

Note. EVT convergence uses the optimiser convergence code recorded in the frozen R handoff. Boundary fits identify solutions that reach the imposed parameter boundary.

Panel B: Formal integrity checks

Source	Check	Passed
R handoff / Phase 1	Crash Main equals Crash 0025	Yes
R handoff / Phase 1	EVT thresholds ordered $1\% \leq 2.5\% \leq 5\%$	Yes
R handoff / Phase 1	EVT shape parameter varies	Yes
R handoff / Phase 1	EVT scale parameter varies	Yes
R handoff / Phase 1	Primary sectors ≥ 3	Yes
Phase 1 final	Corrected R EVT parameter variation passed	Yes
Phase 1 final	R Crash Main equals 2.5% EVT label	Yes
Phase 1 final	R EVT thresholds correctly ordered	Yes
Phase 1 final	No Test dates used in temporal CV	Yes
Phase 1 final	Final stable Hawkes parameters finite	Yes
Phase 1 final	Final stable Hawkes parameters non-negative	Yes
Phase 1 final	Final stable network below branching guard	Yes
Phase 1 final	Final graph source resolved without Test selection	Yes
Phase 1 final	Validation targets contain both classes	Yes
Phase 1 final	Test targets contain both classes	Yes

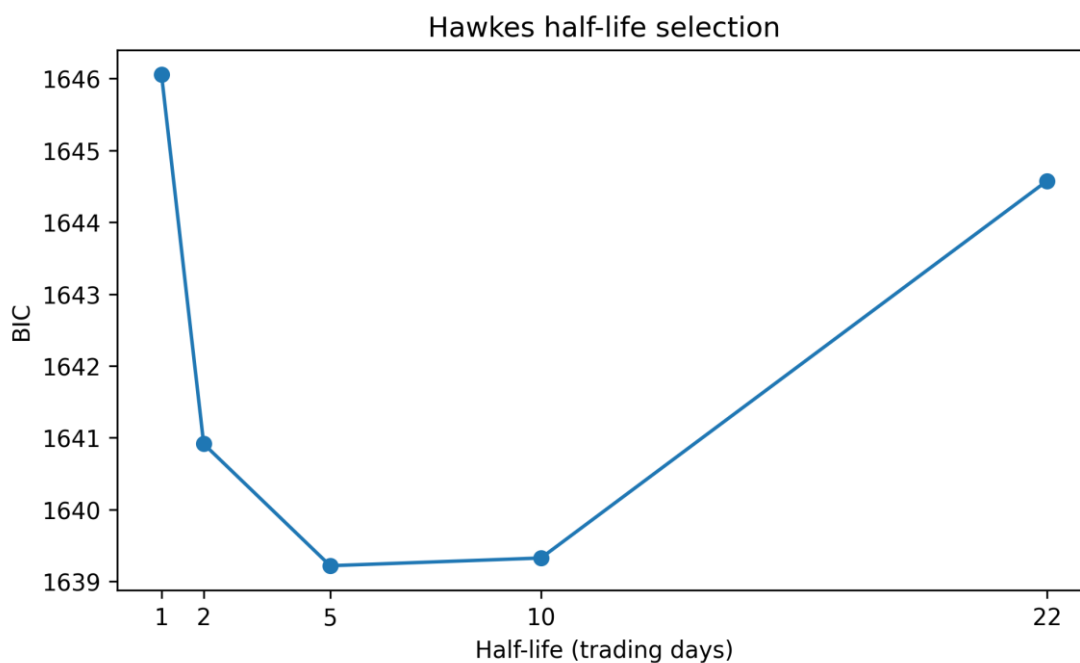
Note. All checks had to pass before the Hawkes and forecasting phases proceeded.

Standardised-innovation standard deviations range from 0.958 to 1.017 across sectors, remaining close to unit scale. All 535 EVT refits converge, and none reaches the imposed parameter boundary. The formal checks also confirm threshold ordering, variation in the EVT parameters, consistency of the frozen 2.5% crash label, and exclusion of Test dates from temporal cross-validation.

SI C. Hawkes network estimation and stability

The Hawkes analysis selects persistence and sparsity using training data only and evaluates edge stability through block subsampling. The first step compares candidate decay specifications using the training-sample Bayesian information criterion (BIC). Figure S1 presents the half-life comparison, and Table S4 reports the full selection and stability diagnostics.

Figure S1. Hawkes half-life selection by training-sample BIC



Note. Lower BIC indicates a better fit.

The five-day specification records the lowest BIC at 1639.22, while the 10-day specification follows closely at 1639.33. The small difference supports persistence over roughly five to ten trading days rather than a precisely identified five-day decay.

Table S4 documents the temporal cross-validation, penalty selection, and stability checks used to determine the frozen Hawkes graph.

Table S4. Hawkes selection and stability diagnostics

Panel A: Half-life comparison

Half-life	Decay	Train log-likelihood	N	Parameters	AIC	BIC	Delta BIC
5	0.8706	-637.7039	5780	42	1359.4079	1639.2186	0.0000
10	0.9330	-637.7578	5780	42	1359.5155	1639.3262	0.1077
2	0.7071	-638.5520	5780	42	1361.1040	1640.9147	1.6962
22	0.9690	-640.3827	5780	42	1364.7653	1644.5760	5.3575
1	0.5000	-641.1220	5780	42	1366.2439	1646.0546	6.8360

Panel B: Temporal cross-validation folds

Fold	Fit end	Validation start	Validation end	Fit dates	Validation dates
1	2020-10-13	2020-10-14	2021-05-18	1048	149
2	2021-05-18	2021-05-19	2021-12-20	1197	149
3	2021-12-20	2021-12-21	2022-07-29	1346	151

Panel C: Temporal-CV penalty grid

Half-life	Lambda	Mean validation Log Score	SD Log Score	Mean cross edges	Folds	SE Log Score
2	0.0	0.3738	0.0821	17.6667	3	0.0474
2	0.005	0.3744	0.0805	17.3333	3	0.0464
2	0.01	0.3752	0.0787	15.6667	3	0.0454
2	0.02	0.3765	0.0754	13.6667	3	0.0435
2	0.05	0.3823	0.0672	9.0000	3	0.0388
2	0.1	0.3860	0.0632	4.6667	3	0.0365
2	0.2	0.3888	0.0575	0.6667	3	0.0332
2	0.5	0.3890	0.0559	0.0000	3	0.0323
5	0.0	0.3930	0.0940	16.3333	3	0.0543
5	0.005	0.3920	0.0928	16.0000	3	0.0536
5	0.01	0.3913	0.0916	15.3333	3	0.0529
5	0.02	0.3906	0.0894	13.6667	3	0.0516
5	0.05	0.3809	0.0787	10.3333	3	0.0454
5	0.1	0.3791	0.0697	4.6667	3	0.0403
5	0.2	0.3810	0.0638	2.0000	3	0.0369
5	0.5	0.3849	0.0575	0.0000	3	0.0332
10	0.0	0.4108	0.0938	16.3333	3	0.0542
10	0.005	0.4099	0.0949	16.3333	3	0.0548
10	0.01	0.4093	0.0958	16.0000	3	0.0553
10	0.02	0.4003	0.0910	15.0000	3	0.0525
10	0.05	0.3898	0.0852	12.6667	3	0.0492
10	0.1	0.3840	0.0802	5.3333	3	0.0463
10	0.2	0.3781	0.0682	3.3333	3	0.0394
10	0.5	0.3828	0.0581	0.0000	3	0.0335

Note. Mean cross edges exclude the diagonal self-excitation terms.

Panel D: Lambda-min and one-SE selections

Half-life	Lambda-min	Minimum CV Log Score	SE at minimum	One-SE limit	Lambda-1SE	Primary penalty	Robustness penalty	Mean cross edges: min	Mean cross edges: 1SE
2	0.0	0.3738	0.0474	0.4212	0.5	0.0	0.5	17.6667	0.0000
5	0.1	0.3791	0.0403	0.4193	0.5	0.1	0.5	4.6667	0.0000
10	0.2	0.3781	0.0394	0.4175	0.5	0.2	0.5	3.3333	0.0000

Panel E: stable-edge counts by half-life

Half-life	Selected lambda	Stable cross edges	Mean cross-edge stability	Maximum cross-edge stability
2	0.0	13	0.505	1.000
5	0.1	3	0.101	0.990
10	0.2	3	0.130	0.840

Panel F: stable-edge overlap across half-lives

Half-life 1	Half-life 2	Edges 1	Edges 2	Common edges	Jaccard
2	5.0	13	3	2	0.143
2	10.0	13	3	2	0.143
5	10.0	3	3	3	1.000

Note. The five-day and 10-day specifications retain the same three stable cross-sector edges.

At the five-day half-life, the lambda-min rule retains three stable cross-sector links, whereas the more conservative one-standard-error penalty removes all cross-sector links. The 10-day specification retains the same three-edge topology as the five-day model, producing a Jaccard overlap of 1.00. These results support a sparse and stable transmission structure while preserving Test-period isolation during network selection.

Table S5 reports all source-to-recipient pairs from the five-day stability exercise and evaluates the standalone forecasting performance of the retained Hawkes specification.

Table S5. Hawkes edge stability and parameter estimates

Panel A: all five-day source-recipient pairs

Source sector	Recipient sector	Self-edge	Selection frequency	Mean coefficient	Stable at 60%	Train a_{ij}	Train+Validat ion a_{ij}
Banking	Banking	Yes	0.9200	0.0317	Yes	0.0349	0.0348
Commercial and services	Banking	No	0.4100	0.0036	No	0.0000	0.0000
Energy and Petroleum	Banking	No	0.0000	0.0000	No	0.0000	0.0000
Insurance	Banking	No	0.5500	0.0042	No	0.0000	0.0000
Investment	Banking	No	0.6200	0.0078	Yes	0.0239	0.0078
Manufacturing and Allied	Banking	No	0.4400	0.0021	No	0.0000	0.0000
Banking	Commercial and services	No	0.0300	0.0000	No	0.0000	0.0000
Commercial and services	Commercial and services	Yes	0.0100	0.0000	No	0.0000	0.0000
Energy and Petroleum	Commercial and services	No	0.0000	0.0000	No	0.0000	0.0000
Insurance	Commercial and services	No	0.0000	0.0000	No	0.0000	0.0000
Investment	Commercial and services	No	0.1000	0.0005	No	0.0000	0.0000
Manufacturing and Allied	Commercial and services	No	0.0200	0.0001	No	0.0000	0.0000
Banking	Energy and Petroleum	No	0.0700	0.0003	No	0.0000	0.0000
Commercial and services	Energy and Petroleum	No	0.0000	0.0000	No	0.0000	0.0000
Energy and Petroleum	Energy and Petroleum	Yes	0.9900	0.0263	Yes	0.0277	0.0090
Insurance	Energy and Petroleum	No	0.5700	0.0048	No	0.0000	0.0000
Investment	Energy and Petroleum	No	0.0900	0.0003	No	0.0000	0.0000
Manufacturing and Allied	Energy and Petroleum	No	0.0000	0.0000	No	0.0000	0.0000

Source sector	Recipient sector	Self-edge	Selection frequency	Mean coefficient	Stable at 60%	Train a_{ij}	Train+ Validation a_{ij}
Banking	Insurance	No	0.9400	0.0222	Yes	0.0611	0.0578
Commercial and services	Insurance	No	0.3200	0.0011	No	0.0000	0.0000
Energy and Petroleum	Insurance	No	0.0000	0.0000	No	0.0000	0.0000
Insurance	Insurance	Yes	0.0000	0.0000	No	0.0000	0.0000
Investment	Insurance	No	0.1300	0.0009	No	0.0000	0.0000
Manufacturing and Allied	Insurance	No	0.0100	0.0000	No	0.0000	0.0000
Banking	Investment	No	0.0100	0.0000	No	0.0000	0.0000
Commercial and services	Investment	No	0.0500	0.0002	No	0.0000	0.0000
Energy and Petroleum	Investment	No	0.0000	0.0000	No	0.0000	0.0000
Insurance	Investment	No	0.0100	0.0000	No	0.0000	0.0000
Investment	Investment	Yes	0.9100	0.0142	Yes	0.0136	0.0000
Manufacturing and Allied	Investment	No	0.0800	0.0002	No	0.0000	0.0000
Banking	Manufacturing and Allied	No	0.1700	0.0005	No	0.0000	0.0000
Commercial and services	Manufacturing and Allied	No	0.0000	0.0000	No	0.0000	0.0000
Energy and Petroleum	Manufacturing and Allied	No	0.0000	0.0000	No	0.0000	0.0000
Insurance	Manufacturing and Allied	No	0.9900	0.0231	Yes	0.0477	0.0343
Investment	Manufacturing and Allied	No	0.4500	0.0029	No	0.0000	0.0000
Manufacturing and Allied	Manufacturing and Allied	Yes	0.0000	0.0000	No	0.0000	0.0099

Note. The 60% stability rule applies to cross-sector edges. Self-excitation terms are retained separately and are not interpreted as cross-sector contagion. The table reports all 36 source-to-recipient pairs across the six primary sectors.

Panel B: standalone pooled Test forecasting

Horizon	Model	N	Events	Brier	Log Score	PR-AUC	ROC-AUC	Brier skill vs expanding history
1	Expanding Historical	2960	66	0.0219	0.1080	0.0200	0.4244	0.0000
5	Expanding Historical	2928	304	0.0941	0.3385	0.0919	0.4288	0.0000
10	Expanding Historical	2890	581	0.1637	0.5110	0.1797	0.4268	0.0000
22	Expanding Historical	2832	1139	0.2462	0.6859	0.3859	0.4431	0.0000
1	Stable Post Selection Hawkes	2960	66	0.0220	0.1093	0.0228	0.4848	-0.0060
5	Stable Post Selection Hawkes	2928	304	0.0958	0.3448	0.0994	0.4719	-0.0178
10	Stable Post Selection Hawkes	2890	581	0.1663	0.5193	0.1950	0.4739	-0.0160
22	Stable Post Selection Hawkes	2832	1139	0.2533	0.7038	0.3987	0.4763	-0.0289

Note. Brier skill uses the expanding historical probability as the reference; positive skill indicates improvement.

Only Investment-to-Banking, Banking-to-Insurance, and Insurance-to-Manufacturing and Allied satisfy the cross-sector stability rule. The standalone Hawkes model records negative Brier skill at every horizon, meaning that its probability forecasts perform worse than the expanding historical benchmark on the Brier Score. The network therefore provides stronger evidence of directional transmission than standalone forecasting performance, supporting its use as conditioning information in the subsequent graph models.

SI D. Machine-learning and deep-learning implementation, calibration, and five-seed diagnostics

The neural robustness analysis fits six architectures across five fixed seeds. The study calibrates each fitted model on the Validation sample before forming the unweighted ensemble for analysis. Table S6 reports the selected training epochs, calibration parameters, and Test performance across seeds.

Table S6. Primary forecast performance and five-seed neural diagnostics

Panel A: Best epoch by seed and architecture

Seed	Model	Best epoch	Best Validation survival negative log-likelihood
20260901	LSTM Survival	2	0.13772
20260901	Temporal Transformer Survival	1	0.12994
20260901	No-Graph Transformer	3	0.13718
20260901	Random-Graph Transformer	4	0.13543
20260901	Static Hawkes Graph Transformer	3	0.13606

Seed	Model	Best epoch	Best Validation survival negative log-likelihood
20260901	Dynamic Hawkes Graph Transformer	3	0.13679
20260902	LSTM Survival	2	0.13531
20260902	Temporal Transformer Survival	2	0.13357
20260902	No-Graph Transformer	4	0.13339
20260902	Random-Graph Transformer	2	0.13479
20260902	Static Hawkes Graph Transformer	4	0.13431
20260902	Dynamic Hawkes Graph Transformer	4	0.13385
20260903	LSTM Survival	2	0.13683
20260903	Temporal Transformer Survival	1	0.13211
20260903	No-Graph Transformer	3	0.13292
20260903	Random-Graph Transformer	3	0.13253
20260903	Static Hawkes Graph Transformer	3	0.13286
20260903	Dynamic Hawkes Graph Transformer	3	0.13311
20260904	LSTM Survival	2	0.13446
20260904	Temporal Transformer Survival	2	0.13449
20260904	No-Graph Transformer	3	0.13610
20260904	Random-Graph Transformer	3	0.13607
20260904	Static Hawkes Graph Transformer	3	0.13614
20260904	Dynamic Hawkes Graph Transformer	3	0.13607
20260905	LSTM Survival	2	0.13582
20260905	Temporal Transformer Survival	1	0.13354
20260905	No-Graph Transformer	4	0.12926
20260905	Random-Graph Transformer	4	0.13226
20260905	Static Hawkes Graph Transformer	4	0.13350
20260905	Dynamic Hawkes Graph Transformer	4	0.13000

Note. Seed denotes the fixed random-number seed used to initialise model training and assess stochastic robustness. The modelling uses five seeds (20260901–20260905) across all neural architectures, with matched graph variants sharing the same initialisation within each seed.

Panel B: Validation calibration parameters

Seed	Model	Temperature	Bias	Learned graph η
20260901	LSTM Survival	1.2096	-1.3893	
20260901	Temporal Transformer Survival	17.0719	-3.4255	
20260901	No-Graph Transformer	1.4989	-1.8045	0.6931
20260901	Random-Graph Transformer	1.9941	-2.2151	0.6883
20260901	Static Hawkes Graph Transformer	1.9247	-2.1838	0.6837
20260901	Dynamic Hawkes Graph Transformer	1.6078	-1.9189	0.6900
20260902	LSTM Survival	1.1835	-1.2877	
20260902	Temporal Transformer Survival	1.0928	-1.0498	
20260902	No-Graph Transformer	1.8160	-2.0476	0.6931
20260902	Random-Graph Transformer	4.6649	-3.0032	0.6851
20260902	Static Hawkes Graph Transformer	2.0946	-2.2595	0.6963
20260902	Dynamic Hawkes Graph Transformer	1.9062	-2.1262	0.6670
20260903	LSTM Survival	1.5989	-1.9104	
20260903	Temporal Transformer Survival	20.0855	-3.4547	
20260903	No-Graph Transformer	1.0889	-1.0183	0.6931
20260903	Random-Graph Transformer	1.1831	-1.2108	0.7044
20260903	Static Hawkes Graph Transformer	1.0973	-1.0365	0.7041
20260903	Dynamic Hawkes Graph Transformer	1.0780	-0.9994	0.7061
20260904	LSTM Survival	1.0902	-1.0798	
20260904	Temporal Transformer Survival	1.6191	-1.8870	
20260904	No-Graph Transformer	0.9311	-0.6914	0.6931

Seed	Model	Temperature	Bias	Learned graph η
20260904	Random-Graph Transformer	0.9557	-0.7634	0.7030
20260904	Static Hawkes Graph Transformer	0.9444	-0.7317	0.7031
20260904	Dynamic Hawkes Graph Transformer	0.9223	-0.6629	0.7017
20260905	LSTM Survival	1.6682	-1.9574	
20260905	Temporal Transformer Survival	18.2821	-3.4384	
20260905	No-Graph Transformer	1.3773	-1.4576	0.6931
20260905	Random-Graph Transformer	1.1573	-1.1561	0.7036
20260905	Static Hawkes Graph Transformer	1.2056	-1.2870	0.6797
20260905	Dynamic Hawkes Graph Transformer	1.4015	-1.5161	0.6925

Note. Learned graph η is reported for Transformer variants that retain the graph-attention parameter; it is not applicable to the LSTM or temporal Transformer architectures.

Panel C: Test performance mean and standard deviation across seeds

Model	Horizon	Brier mean	Brier SD	Log Score mean	Log Score SD	PR-AUC mean	PR-AUC SD	ROC-AUC mean	ROC-AUC SD	Calibration slope mean	Calibration slope SD
Dynamic Hawkes Graph Transformer	1	0.0218	0.0000	0.1077	0.0007	0.0281	0.0019	0.4898	0.0175	0.8359	0.5064
Dynamic Hawkes Graph Transformer	5	0.0938	0.0003	0.3368	0.0013	0.1138	0.0014	0.4975	0.0074	0.1114	0.4947
Dynamic Hawkes Graph Transformer	10	0.1621	0.0007	0.5061	0.0018	0.2162	0.0031	0.4917	0.0056	-0.0603	0.4675
Dynamic Hawkes Graph Transformer	22	0.2406	0.0009	0.6742	0.0018	0.4460	0.0059	0.5058	0.0083	0.7057	0.4407
LSTM Survival	1	0.0218	0.0000	0.1068	0.0001	0.0328	0.0087	0.5451	0.0282	2.6272	2.5229
LSTM Survival	5	0.0933	0.0004	0.3347	0.0016	0.1220	0.0121	0.5318	0.0040	1.8174	1.5759
LSTM Survival	10	0.1610	0.0010	0.5029	0.0026	0.2259	0.0232	0.5243	0.0092	1.0208	0.8465
LSTM Survival	22	0.2392	0.0022	0.6714	0.0044	0.4484	0.0379	0.5331	0.0200	1.3123	1.1128
No Graph Transformer	1	0.0218	0.0000	0.1078	0.0007	0.0287	0.0022	0.4915	0.0154	0.6730	0.5517
No Graph Transformer	5	0.0938	0.0003	0.3370	0.0013	0.1136	0.0018	0.4966	0.0066	0.1039	0.5233
No Graph Transformer	10	0.1622	0.0007	0.5063	0.0018	0.2151	0.0035	0.4900	0.0048	-0.0730	0.4296
No Graph Transformer	22	0.2406	0.0008	0.6743	0.0017	0.4455	0.0064	0.5036	0.0074	0.7278	0.3698
Random Graph Transformer	1	0.0218	0.0000	0.1076	0.0007	0.0275	0.0020	0.4946	0.0226	0.6290	0.6314
Random Graph Transformer	5	0.0938	0.0003	0.3367	0.0014	0.1115	0.0029	0.4937	0.0073	-1.5197	3.2876
Random Graph Transformer	10	0.1621	0.0004	0.5061	0.0012	0.2111	0.0040	0.4865	0.0088	-2.0808	4.4492
Random Graph Transformer	22	0.2409	0.0009	0.6750	0.0019	0.4399	0.0076	0.5018	0.0088	0.8197	0.1601
Static Hawkes Graph Transformer	1	0.0218	0.0000	0.1077	0.0006	0.0272	0.0020	0.4914	0.0154	0.8322	0.5115
Static Hawkes Graph Transformer	5	0.0938	0.0003	0.3370	0.0013	0.1119	0.0028	0.4952	0.0061	-0.0739	0.7581
Static Hawkes Graph Transformer	10	0.1622	0.0006	0.5063	0.0017	0.2128	0.0024	0.4889	0.0078	-0.3336	0.9141
Static Hawkes Graph Transformer	22	0.2408	0.0010	0.6746	0.0020	0.4407	0.0066	0.5019	0.0119	0.4461	0.9460
Temporal Transformer Survival	1	0.0219	0.0001	0.1076	0.0007	0.0261	0.0044	0.5135	0.0364	0.4192	0.9341
Temporal Transformer Survival	5	0.0940	0.0007	0.3371	0.0018	0.1135	0.0063	0.5013	0.0255	-1.3386	3.8314
Temporal Transformer Survival	10	0.1624	0.0007	0.5069	0.0023	0.2153	0.0110	0.4981	0.0278	0.5863	1.6421
Temporal Transformer Survival	22	0.2424	0.0003	0.6796	0.0029	0.4301	0.0276	0.5094	0.0327	0.5969	2.1162

Note. The statistics summarise five independently calibrated seed forecasts for each neural architecture. The final ensemble averages the calibrated probabilities rather than the evaluation metrics.

The five-seed diagnostics show that training and probability-loss performance remain broadly stable, while calibration is more sensitive to stochastic initialisation. Panel A shows that early stopping selects similar epochs across seeds, with the LSTM consistently stopping at epoch 2 and the Transformer variants generally stopping between epochs 1 and 4. Panel B indicates that the graph-attention parameter remains close to 0.69 across graph variants and seeds, whereas calibration adjustments vary more noticeably, particularly for the

temporal Transformer. Panel C confirms that Brier Score and Log Score dispersion is small across seeds for most architectures. Among the neural models, the LSTM records the lowest mean Brier Score at 5, 10, and 22 days and remains comparable at one day, while calibration slopes vary much more strongly across several Transformer specifications. These results show that a single neural initialisation does not drive the main probability-loss comparisons, although calibration remains less stable than Brier and Log Score performance.

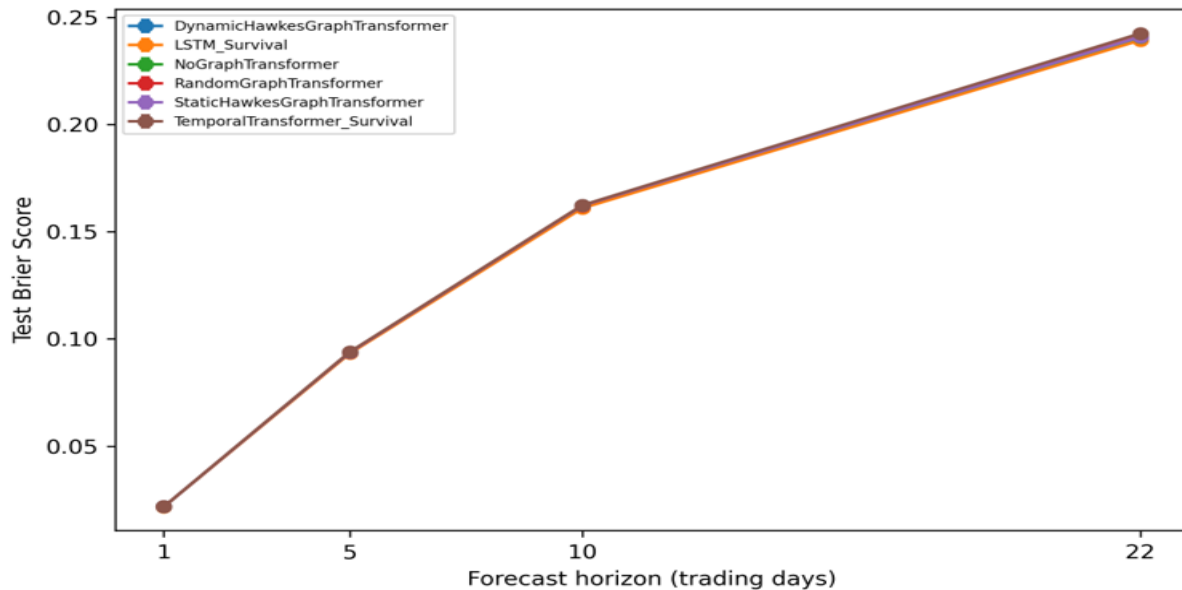
Table S6. Panel D. Primary locked Test ensemble and benchmark performance

Horizon	Model	N	Events	Brier Score	Log Score	PR-AUC	ROC-AUC
1	XGBoost	2960	66	0.021779	0.106631	0.059893	0.627149
1	LSTM Survival	2960	66	0.021793	0.106716	0.029277	0.547682
1	Dynamic Hawkes Graph Transformer	2960	66	0.021840	0.107627	0.027873	0.480629
1	No-Graph Transformer	2960	66	0.021842	0.107659	0.028284	0.482027
1	Expanding Historical	2960	66	0.021853	0.107971	0.019952	0.424434
5	XGBoost	2928	304	0.092055	0.328154	0.163226	0.610452
5	LSTM Survival	2928	304	0.093225	0.334315	0.121852	0.533321
5	Dynamic Hawkes Graph Transformer	2928	304	0.093759	0.336732	0.112594	0.494043
5	No-Graph Transformer	2928	304	0.093785	0.336847	0.112881	0.494125
5	Expanding Historical	2928	304	0.094088	0.338503	0.091882	0.428813
10	LSTM Survival	2890	581	0.160742	0.502273	0.234676	0.529037
10	Dynamic Hawkes Graph Transformer	2890	581	0.162021	0.505832	0.213083	0.488779
10	No-Graph Transformer	2890	581	0.162085	0.506008	0.212261	0.488159
10	XGBoost	2890	581	0.162736	0.507789	0.218483	0.524906
10	Expanding Historical	2890	581	0.163728	0.511004	0.179704	0.426786
22	XGBoost	2832	1139	0.234281	0.660502	0.498542	0.598208
22	LSTM Survival	2832	1139	0.238618	0.670199	0.470794	0.541208
22	Dynamic Hawkes Graph Transformer	2832	1139	0.240402	0.673858	0.443369	0.505350
22	No-Graph Transformer	2832	1139	0.240465	0.673986	0.442481	0.504036
22	Expanding Historical	2832	1139	0.246176	0.685901	0.385884	0.443052

Note. Panel D reproduces the locked Phase 2.2 pooled Test forecasts underlying Table 3 of the main manuscript. Neural-model metrics are calculated from the five-seed ensemble probabilities, whereas XGBoost and the expanding historical benchmark are deterministic under the frozen primary specification. Panel C instead reports the mean and standard deviation of metrics calculated separately across the five neural fits before aggregation.

The five-seed results show how Brier Score variation changes across the four forecast horizons, as illustrated in Figure S2.

Figure S2. Five-seed neural Test Brier Score mean and standard deviation



Note. Error bars show one standard deviation across the five fitted seeds for each neural architecture.

The models remain closely clustered at the shorter horizons, while differences become more visible as the forecast horizon lengthens. Seed-level variation remains modest for most Brier Score comparisons, and no graph architecture achieves uniform dominance across all four horizons.

SI E. Graph placebos and ablations

The graph-placebo analysis tests whether forecasting differences arise from the estimated Hawkes topology or from graph conditioning more generally. Table S7 documents the matched graph variants, training-derived magnitude scaling, preservation of edge-weight variation, graph-state features, and random cross-sector supports used across seeds.

Table S7. Graph placebo design and random supports

Panel A: Graph-ablation design

Graph variant	Description	Cross edges	Row normalized	Explicit graph-state features
No-Graph	Zero econometric prior; graph attention remains data-learned; graph-state summaries are zero.	0	NO	ZERO
Random Static Graph	Random directed prior with Hawkes cross-edge density; same Train-derived magnitude scale.	3	NO	YES
Static Hawkes Graph	Time-invariant stability-selected Hawkes coefficient prior; same Train-derived magnitude scale.	3	NO	YES
Dynamic Hawkes Graph	Time-varying stability-selected Hawkes excitation prior; absolute excitation magnitude preserved.	3	NO	YES

Note. The no-graph model retains the same neural architecture but sets the econometric graph prior and graph-state summaries to zero.

Panel B: Train-derived magnitude-scaling audit

Scaling audit item	Value
Scaling method	$\log_{10}(A/q_{\text{train}})/\log(2)$
Train-only graph quantile	0.99
Train-only q reference	0.061189
Global clip	3
Row normalisation used	NO
Positive Train raw minimum	0.00000000000000000178
Positive Train raw median	0.000542
Positive Train raw q90	0.022603
Positive Train raw q99	0.061189
Positive Train raw maximum	0.164838
Scaled Train positive median	0.012725
Scaled Train positive q90	0.453544
Scaled Train positive q99	1
Scaled Train positive maximum	1.885146

Note. The same Train-derived transform applies to Train, Validation, Test, static, and random graph weights.

Panel C: Edge-magnitude preservation

From sector	To sector	Raw min	Raw median	Raw max	Scaled min	Scaled median	Scaled max	Raw-scaled correlation	Unique scaled values
Investment	Banking	0.0000	0.0000	0.0466	0.0000	0.0002	0.8169	0.9978	874
Banking	Insurance	0.0000	0.0000	0.1648	0.0000	0.0000	1.8851	0.9866	741
Insurance	Manufacturing and Allied	0.0000	0.0001	0.1193	0.0000	0.0019	1.5603	0.9928	951

Note. High raw-scaled correlations confirm that the magnitude transform preserves ordering and variation rather than collapsing active edges to a common value.

Panel D: Dynamic node-level graph-state features

Graph-state feature	Train mean	Train SD	Train maximum
Hawkes Incoming Risk	0.0535	0.1720	1.8851
Hawkes Outgoing Risk	0.0535	0.1720	1.8851
Hawkes Max Incoming Risk	0.0535	0.1720	1.8851

Note. The three graph-state summaries enter as node features in addition to the soft attention prior.

Panel E: Random-graph supports by seed

Seed	Edge	From sector	To sector	Train placebo weight	Train+Validation placebo weight	Preserves self-excitation
20260901	1	Investment	Manufacturing and Allied	0.06110	0.05782	YES
20260901	2	Manufacturing and Allied	Insurance	0.02390	0.00781	YES
20260901	3	Insurance	Banking	0.04768	0.03427	YES
20260902	1	Energy and Petroleum	Banking	0.02390	0.00781	YES
20260902	2	Manufacturing and Allied	Commercial and services	0.04768	0.03427	YES
20260902	3	Insurance	Manufacturing and Allied	0.06110	0.05782	YES
20260903	1	Investment	Energy and Petroleum	0.02390	0.00781	YES
20260903	2	Commercial and services	Investment	0.04768	0.03427	YES
20260903	3	Insurance	Manufacturing and Allied	0.06110	0.05782	YES
20260904	1	Investment	Energy and Petroleum	0.06110	0.05782	YES
20260904	2	Energy and Petroleum	Banking	0.02390	0.00781	YES
20260904	3	Insurance	Commercial and services	0.04768	0.03427	YES
20260905	1	Investment	Banking	0.02390	0.00781	YES
20260905	2	Insurance	Commercial and services	0.04768	0.03427	YES
20260905	3	Investment	Energy and Petroleum	0.06110	0.05782	YES

Table S7 shows that the graph ablations differ in network information while preserving a closely matched forecasting architecture. Panel A separates the no-graph, random, static-Hawkes, and dynamic-Hawkes variants, while holding the number of cross-sector edges at three for the graph-based controls. Panel B shows that all graph variants use the same training-derived magnitude transformation without row normalisation, preventing scaling differences from driving the comparison. Panel C further confirms that the transformation preserves the ordering and variation of the retained Hawkes weights, with raw-scaled correlations above 0.98 for all three stable cross-sector links.

Panel D shows that the dynamic specification also incorporates Hawkes-based incoming, outgoing, and maximum incoming risk as node-level state variables in addition to the soft graph prior. Panel E then varies the random cross-sector support across the five seeds while preserving graph density, empirical weight magnitudes, and self-excitation. This design makes the placebo comparison more stringent because the random graph changes the topology without simultaneously changing the number or scale of cross-sector links. The subsequent ablations can therefore distinguish the predictive value of the estimated Hawkes structure from the more general effect of graph conditioning.

Table S8 evaluates the Dynamic Hawkes Graph model against no-graph, random-graph, static-Hawkes, and broader forecasting benchmarks using both paired-seed comparisons and moving-block-bootstrap inference.

Table S8. Graph paired-seed and moving-block-bootstrap comparisons

Panel A: Paired seed summary

Horizon	Comparison	Seeds	Mean Δ Brier	SD Δ Brier	Dynamic better share: Brier	Mean Δ Log Score	SD Δ Log Score	Dynamic better share: Log Score	Mean Δ PR-AUC	Mean Δ ROC-AUC
1	No Graph Transformer	5	-0.00000	0.00000	0.80000	-0.00004	0.00006	0.60000	-0.00056	-0.00170
1	Random Graph Transformer	5	0.00001	0.00001	0.40000	0.00009	0.00025	0.40000	0.00059	-0.00481

Horizon	Comparison	Seeds	Mean Δ Brier	SD Δ Brier	Dynamic better share: Brier	Mean Δ Log Score	SD Δ Log Score	Dynamic better share: Log Score	Mean Δ PR-AUC	Mean Δ ROC-AUC
1	Static Hawkes Graph Transformer	5	0.00000	0.00001	0.40000	0.00002	0.00008	0.40000	0.00090	-0.00162
5	No Graph Transformer	5	-0.00003	0.00002	1.00000	-0.00011	0.00009	1.00000	0.00020	0.00087
5	Random Graph Transformer	5	0.00002	0.00029	0.60000	0.00010	0.00126	0.60000	0.00237	0.00376
5	Static Hawkes Graph Transformer	5	-0.00004	0.00008	0.40000	-0.00017	0.00032	0.40000	0.00193	0.00230
10	No Graph Transformer	5	-0.00006	0.00005	1.00000	-0.00017	0.00015	1.00000	0.00107	0.00178
10	Random Graph Transformer	5	-0.00001	0.00073	0.80000	-0.00003	0.00201	0.60000	0.00508	0.00520
10	Static Hawkes Graph Transformer	5	-0.00009	0.00015	0.60000	-0.00024	0.00040	0.60000	0.00342	0.00282
22	No Graph Transformer	5	-0.00005	0.00011	0.60000	-0.00010	0.00023	0.60000	0.00055	0.00220
22	Random Graph Transformer	5	-0.00036	0.00148	0.80000	-0.00072	0.00300	0.80000	0.00612	0.00403
22	Static Hawkes Graph Transformer	5	-0.00018	0.00021	0.80000	-0.00036	0.00042	0.80000	0.00529	0.00395

Note. Dynamic-minus-benchmark differences are reported. Negative Brier and Log Score differences favour the Dynamic Hawkes Graph model.

Panel B: Ensemble block-bootstrap comparisons

Horizon	Benchmark	Δ Brier	Brier CI 2.5%	Brier CI 97.5%	P (dynamic better)	Δ Log Score	Log Score CI 2.5%	Log Score CI 97.5%	P (dynamic better)
1	LSTM Survival	0.00005	-0.00001	0.00009	0.04800	0.00091	-0.00020	0.00185	0.05600
1	Temporal Transformer Survival	-0.00001	-0.00005	0.00002	0.59000	0.00006	-0.00034	0.00045	0.30400
1	No Graph Transformer	-0.00000	-0.00000	0.00000	0.90800	-0.00003	-0.00007	0.00001	0.90000
1	Random Graph Transformer	0.00001	-0.00000	0.00002	0.05000	0.00014	-0.00002	0.00029	0.05200
1	Static Hawkes Graph Transformer	0.00000	-0.00000	0.00000	0.15000	0.00002	-0.00002	0.00006	0.15000
1	XGBoost	0.00006	-0.00016	0.00020	0.39600	0.00101	-0.00414	0.00526	0.39200
1	Fixed Historical	-0.00004	-0.00014	0.00004	0.81000	-0.00087	-0.00308	0.00090	0.83200
1	Expanding Historical	-0.00001	-0.00009	0.00005	0.65800	-0.00034	-0.00204	0.00098	0.68200
1	Stable Hawkes	-0.00014	-0.00028	-0.00000	0.98200	-0.00168	-0.00413	0.00067	0.91400
5	LSTM Survival	0.00054	0.00017	0.00092	0.00200	0.00244	0.00073	0.00413	0.00200
5	Temporal Transformer Survival	-0.00006	-0.00048	0.00037	0.40800	0.00010	-0.00111	0.00158	0.25600
5	No Graph Transformer	-0.00003	-0.00006	0.00001	0.94200	-0.00012	-0.00025	0.00003	0.94200
5	Random Graph Transformer	0.00005	-0.00000	0.00010	0.03200	0.00021	-0.00001	0.00044	0.04600
5	Static Hawkes Graph Transformer	-0.00003	-0.00009	0.00002	0.89600	-0.00015	-0.00039	0.00009	0.88600
5	XGBoost	0.00173	-0.00079	0.00396	0.10200	0.00870	-0.00539	0.02204	0.12400
5	Fixed Historical	-0.00072	-0.00184	0.00036	0.89600	-0.00383	-0.00977	0.00184	0.89400
5	Expanding Historical	-0.00031	-0.00126	0.00056	0.72200	-0.00170	-0.00655	0.00272	0.73600
5	Stable Hawkes	-0.00201	-0.00410	0.00033	0.95800	-0.00806	-0.01801	0.00282	0.93400
10	LSTM Survival	0.00127	0.00025	0.00216	0.00400	0.00353	0.00067	0.00602	0.00400
10	Temporal Transformer Survival	-0.00005	-0.00059	0.00066	0.43000	-0.00000	-0.00141	0.00186	0.37800
10	No Graph Transformer	-0.00007	-0.00016	0.00002	0.93600	-0.00018	-0.00043	0.00005	0.93800
10	Random Graph Transformer	-0.00001	-0.00016	0.00015	0.55400	-0.00003	-0.00046	0.00042	0.56800
10	Static Hawkes Graph Transformer	-0.00009	-0.00025	0.00008	0.85800	-0.00023	-0.00071	0.00023	0.84600
10	XGBoost	-0.00072	-0.00237	0.00110	0.77200	-0.00198	-0.00656	0.00304	0.76800
10	Fixed Historical	-0.00313	-0.00630	-0.00023	0.98600	-0.00974	-0.01993	-0.00048	0.98200
10	Expanding Historical	-0.00168	-0.00436	0.00079	0.90200	-0.00510	-0.01329	0.00239	0.89800
10	Stable Hawkes	-0.00429	-0.00947	0.00063	0.94200	-0.01339	-0.02951	0.00411	0.93000
22	LSTM Survival	0.00162	0.00007	0.00358	0.01400	0.00332	0.00017	0.00733	0.01400
22	Temporal Transformer Survival	-0.00100	-0.00423	0.00138	0.83000	-0.00202	-0.00867	0.00286	0.83000
22	No Graph Transformer	-0.00004	-0.00021	0.00008	0.81800	-0.00009	-0.00043	0.00016	0.82200
22	Random Graph Transformer	-0.00010	-0.00117	0.00065	0.70000	-0.00018	-0.00239	0.00135	0.69400
22	Static Hawkes Graph Transformer	-0.00006	-0.00057	0.00023	0.77200	-0.00011	-0.00117	0.00049	0.76000
22	XGBoost	0.00432	-0.00432	0.01768	0.11000	0.00939	-0.00946	0.03794	0.11000
22	Fixed Historical	-0.01072	-0.01566	-0.00240	1.00000	-0.02288	-0.03332	-0.00483	1.00000
22	Expanding Historical	-0.00588	-0.01025	-0.00072	0.99000	-0.01226	-0.02138	-0.00169	0.98800
22	Stable Hawkes	-0.01619	-0.02506	0.00114	0.95800	-0.03800	-0.05760	0.00219	0.96000

Note. Bootstrap inference uses 500 replications with 22-trading-day moving blocks to preserve short-run temporal dependence in the Test sample. P (dynamic better) is the proportion of bootstrap replications in which the Dynamic Hawkes Graph model records a lower Brier Score or Log Score than the comparison model. Values closer to 1 therefore indicate stronger bootstrap evidence in favour of the Dynamic Hawkes Graph model.

The dynamic graph improves on the no-graph control most consistently at 5 and 10 days, but the five-seed ensemble confidence intervals include zero. Comparisons with random and static graphs remain similarly small, so the evidence does not identify the Hawkes topology as uniquely superior.

Table S9 moves the same comparison to the sector-by-horizon level to determine whether pooled graph effects mask important cross-sectional heterogeneity.

Table S9. Sector-by-horizon Dynamic Hawkes Graph heterogeneity

Sector	Horizon	N	Events	Event rate (%)	Brier	Log Score	PR-AUC	ROC-AUC	Calibration slope	Brier skill vs no graph	Log Score improvement vs no graph
Banking	1	494	15	3.04	0.0294	0.1361	0.0412	0.6219	0.9750	-0.000071	-0.0000
Commercial and services	1	492	10	2.03	0.0200	0.1006	0.0349	0.5390	1.0902	0.000150	0.0001
Energy and Petroleum	1	494	6	1.21	0.0123	0.0716	0.0154	0.5376	1.2248	0.000429	0.0001
Insurance	1	494	10	2.02	0.0199	0.1006	0.0438	0.5758	6.8715	0.000100	0.0000
Investment	1	494	10	2.02	0.0199	0.1005	0.0580	0.6779	6.8968	0.000216	0.0001
Manufacturing and Allied	1	492	15	3.05	0.0296	0.1365	0.0537	0.6892	6.5533	-0.000057	-0.0000
Banking	5	490	64	13.06	0.1136	0.3877	0.1526	0.5542	6.1439	-0.000076	-0.0000
Commercial and services	5	484	46	9.50	0.0871	0.3193	0.1536	0.5622	6.3715	0.000467	0.0002
Energy and Petroleum	5	490	30	6.12	0.0617	0.2535	0.0706	0.5408	2.3204	0.001271	0.0004
Insurance	5	490	50	10.20	0.0926	0.3342	0.1565	0.6045	7.5447	-0.000015	0.0000
Investment	5	490	47	9.59	0.0884	0.3237	0.1486	0.6486	9.0402	0.000764	0.0003
Manufacturing and Allied	5	484	67	13.84	0.1194	0.4026	0.2152	0.6500	7.3380	-0.000170	-0.0001
Banking	10	485	119	24.54	0.1857	0.5587	0.2578	0.5291	1.8107	-0.000203	-0.0001
Commercial and services	10	474	83	17.51	0.1477	0.4737	0.2423	0.5549	6.8009	0.000532	0.0002
Energy and Petroleum	10	485	60	12.37	0.1197	0.4109	0.1508	0.5222	6.1810	0.001514	0.0005
Insurance	10	485	95	19.59	0.1600	0.5016	0.2371	0.5717	4.5520	0.000408	0.0002
Investment	10	485	92	18.97	0.1566	0.4941	0.2795	0.6681	9.2295	0.001007	0.0004
Manufacturing and Allied	10	476	132	27.73	0.2029	0.5970	0.4273	0.6889	7.3643	-0.000296	-0.0002
Banking	22	473	213	45.03	0.2498	0.6928	0.4858	0.5285	3.6368	-0.000241	-0.0001
Commercial and services	22	460	156	33.91	0.2315	0.6559	0.3402	0.4657	-2.2863	0.000266	0.0001
Energy and Petroleum	22	478	132	27.62	0.2192	0.6310	0.3075	0.5006	2.4383	0.001115	0.0005
Insurance	22	473	203	42.92	0.2437	0.6806	0.4830	0.5817	4.7273	0.000501	0.0002
Investment	22	473	191	40.38	0.2379	0.6689	0.5932	0.7278	10.0000	0.000190	0.0001
Manufacturing and Allied	22	475	244	51.37	0.2602	0.7138	0.7117	0.7173	10.0000	-0.000149	-0.0001

Note. Positive Brier skill and positive Log Score improvement indicate that the Dynamic Hawkes Graph model improves on the no-graph Transformer for that sector-horizon cell.

Graph contributions vary in sign and magnitude across sectors and forecast horizons. At 10 days, the Dynamic Hawkes Graph model records its largest positive Brier-skill contributions relative to the no-graph Transformer for Energy and Petroleum and Investment, whereas Banking and Manufacturing and Allied show small negative contributions. Similar mixed sectoral effects occur at the other horizons, indicating that pooled comparisons conceal differences in the incremental value of graph information. The results therefore support H5 by showing that the forecasting contribution of graph conditioning depends on both sector and horizon rather than producing a uniform market-wide improvement.

SI F. Component and crash-definition robustness

Component ablations assess the incremental forecasting contribution of GJR-GARCH information, the broader volatility block, EVT/tail variables, liquidity, breadth and dispersion, and internal market-state information. Table S10 reports pooled Test performance and paired moving-block-bootstrap comparisons for XGBoost and the dynamic ridge-logit benchmark.

Tables S10–S14 report results from the separately frozen Phase 3 robustness exercise. These tables use matched within-phase refits for component-ablation and crash-definition comparisons and therefore do not reproduce the primary Phase 2.2 model-comparison forecasts reported in Table 3 of the main manuscript and Table S6, Panel D. Consequently, interpret all robustness comparisons within their corresponding Phase 3 specification.

Table S10. Component-ablation robustness**Panel A: Pooled Test performance by component specification**

Model family	Variant	Horizon	N	Events	Brier	Log Score	PR-AUC	ROC-AUC	Calibration slope
XGBoost	Full	1	2960	66	0.0218	0.1064	0.0480	0.6251	0.7563
XGBoost	Full	5	2928	304	0.0919	0.3269	0.1562	0.6190	1.1581
XGBoost	Full	10	2890	581	0.1618	0.5052	0.2403	0.5572	1.8694
XGBoost	Full	22	2832	1139	0.2370	0.6667	0.4826	0.5719	0.9193
XGBoost	No GARCH volatility	1	2960	66	0.0218	0.1066	0.0316	0.6265	0.7180
XGBoost	No GARCH volatility	5	2928	304	0.0923	0.3292	0.1573	0.6099	0.9363
XGBoost	No GARCH volatility	10	2890	581	0.1577	0.4928	0.2771	0.5957	1.4774
XGBoost	No GARCH volatility	22	2832	1139	0.2370	0.6663	0.4444	0.5798	0.9250
XGBoost	No Volatility Group	1	2960	66	0.0217	0.1057	0.0410	0.6168	0.8828
XGBoost	No Volatility Group	5	2928	304	0.0920	0.3269	0.1588	0.6332	1.1641
XGBoost	No Volatility Group	10	2890	581	0.1628	0.5079	0.2010	0.5000	1.1334
XGBoost	No Volatility Group	22	2832	1139	0.2370	0.6664	0.4452	0.5870	1.2156
XGBoost	No Tail EVT	1	2960	66	0.0218	0.1062	0.0455	0.6053	0.8006
XGBoost	No Tail EVT	5	2928	304	0.0921	0.3284	0.1554	0.5977	1.0347
XGBoost	No Tail EVT	10	2890	581	0.1627	0.5075	0.2224	0.5405	1.1623
XGBoost	No Tail EVT	22	2832	1139	0.2350	0.6628	0.5003	0.5939	1.5049
XGBoost	No Liquidity	1	2960	66	0.0218	0.1056	0.0388	0.6550	1.1313
XGBoost	No Liquidity	5	2928	304	0.0920	0.3270	0.1585	0.6262	1.2718
XGBoost	No Liquidity	10	2890	581	0.1601	0.5005	0.2421	0.5543	0.7838
XGBoost	No Liquidity	22	2832	1139	0.2402	0.6733	0.4340	0.5563	0.5209
XGBoost	No breadth/dispersion	1	2960	66	0.0218	0.1063	0.0389	0.6306	0.8664
XGBoost	No breadth/dispersion	5	2928	304	0.0916	0.3243	0.1640	0.6430	1.2718
XGBoost	No breadth/dispersion	10	2890	581	0.1649	0.5133	0.1910	0.4658	-1.2298
XGBoost	No breadth/dispersion	22	2832	1139	0.2401	0.6730	0.4433	0.5478	0.7247
XGBoost	No Market State	1	2960	66	0.0218	0.1064	0.0347	0.6353	0.8520
XGBoost	No Market State	5	2928	304	0.0920	0.3270	0.1546	0.6315	1.0294
XGBoost	No Market State	10	2890	581	0.1628	0.5080	0.2280	0.5138	0.1219
XGBoost	No Market State	22	2832	1139	0.2428	0.6788	0.4341	0.5236	0.3869
Dynamic ridge-logit	Full	1	2960	66	0.0219	0.1201	0.0420	0.6381	0.1479
Dynamic ridge-logit	Full	5	2928	304	0.0979	0.4082	0.1639	0.6283	0.1225
Dynamic ridge-logit	Full	10	2890	581	0.1815	0.6868	0.2681	0.6013	0.0824
Dynamic ridge-logit	Full	22	2832	1139	0.3073	1.1943	0.4737	0.5902	0.0522
Dynamic ridge-logit	No GARCH volatility	1	2960	66	0.0219	0.1213	0.0421	0.6378	0.1452
Dynamic ridge-logit	No GARCH volatility	5	2928	304	0.0970	0.4138	0.1623	0.6176	0.1117
Dynamic ridge-logit	No GARCH volatility	10	2890	581	0.1823	0.7051	0.2571	0.5915	0.0727
Dynamic ridge-logit	No GARCH volatility	22	2832	1139	0.3200	1.2504	0.4650	0.5793	0.0464
Dynamic ridge-logit	No Volatility Group	1	2960	66	0.0219	0.1210	0.0426	0.6257	0.1259
Dynamic ridge-logit	No Volatility Group	5	2928	304	0.0973	0.4198	0.1601	0.6144	0.0948
Dynamic ridge-logit	No Volatility Group	10	2890	581	0.1819	0.7126	0.2548	0.5899	0.0638
Dynamic ridge-logit	No Volatility Group	22	2832	1139	0.3229	1.2690	0.4649	0.5799	0.0426
Dynamic ridge-logit	No Tail EVT	1	2960	66	0.0220	0.1118	0.0442	0.6565	0.2179
Dynamic ridge-logit	No Tail EVT	5	2928	304	0.1028	0.3805	0.1663	0.6219	0.2351
Dynamic ridge-logit	No Tail EVT	10	2890	581	0.2235	0.6917	0.2816	0.6090	0.1750
Dynamic ridge-logit	No Tail EVT	22	2832	1139	0.3108	1.1482	0.4803	0.6131	0.0983
Dynamic ridge-logit	No Liquidity	1	2960	66	0.0218	0.1206	0.0449	0.6395	0.1489
Dynamic ridge-logit	No Liquidity	5	2928	304	0.0979	0.4074	0.1654	0.6369	0.1291
Dynamic ridge-logit	No Liquidity	10	2890	581	0.1828	0.6808	0.2628	0.6040	0.0796
Dynamic ridge-logit	No Liquidity	22	2832	1139	0.2968	1.1612	0.4808	0.5981	0.0512
Dynamic ridge-logit	No breadth/dispersion	1	2960	66	0.0219	0.1220	0.0417	0.6299	0.1411
Dynamic ridge-logit	No breadth/dispersion	5	2928	304	0.0973	0.4090	0.1666	0.6359	0.1219
Dynamic ridge-logit	No breadth/dispersion	10	2890	581	0.1766	0.6811	0.2683	0.6030	0.0819
Dynamic ridge-logit	No breadth/dispersion	22	2832	1139	0.3113	1.2316	0.4797	0.5969	0.0482
Dynamic ridge-logit	No Market State	1	2960	66	0.0219	0.1237	0.0398	0.6279	0.1373
Dynamic ridge-logit	No Market State	5	2928	304	0.0964	0.4119	0.1678	0.6449	0.1290
Dynamic ridge-logit	No Market State	10	2890	581	0.1780	0.6487	0.2807	0.6198	0.0875
Dynamic ridge-logit	No Market State	22	2832	1139	0.2745	1.0237	0.5070	0.6154	0.0722

Table S10. Component-ablation robustness
Panel B: Moving-block-bootstrap full-versus-ablation comparisons

Model family	Ablation	Horizon	Full-ablation Δ Brier	Brier CI 2.5%	Brier CI 97.5%	P (full better): Brier	Full-ablation Δ Log Score	Log CI 2.5%	Log CI 97.5%	P (full better): Log
XGBoost	No GARCH volatility	1	-0.000037	-0.000077	+0.000025	0.780	-0.00023	-0.00121	+0.00082	0.638
XGBoost	No GARCH volatility	5	-0.000342	-0.001009	+0.000176	0.876	-0.00231	-0.00661	+0.00107	0.892
XGBoost	No GARCH volatility	10	+0.004153	+0.000622	+0.008635	0.010	+0.01241	+0.00214	+0.02596	0.012
XGBoost	No GARCH volatility	22	-0.000086	-0.009883	+0.012845	0.448	+0.00026	-0.02015	+0.02758	0.438
XGBoost	No volatility group	1	+0.000019	-0.000031	+0.000086	0.152	+0.00070	-0.00083	+0.00235	0.188
XGBoost	No volatility group	5	-0.000024	-0.000596	+0.000378	0.604	-0.00000	-0.00373	+0.00243	0.584
XGBoost	No volatility group	10	-0.000973	-0.002271	+0.000079	0.960	-0.00271	-0.00616	+0.00016	0.962
XGBoost	No volatility group	22	+0.000807	-0.010489	+0.011240	0.490	+0.00196	-0.02141	+0.02367	0.478
XGBoost	No tail/EVT	1	-0.000004	-0.000045	+0.000071	0.312	+0.00017	-0.00104	+0.00153	0.370
XGBoost	No tail/EVT	5	-0.000188	-0.000981	+0.000511	0.776	-0.00144	-0.00625	+0.00241	0.792
XGBoost	No tail/EVT	10	-0.000931	-0.003009	+0.000924	0.838	-0.00236	-0.00778	+0.00248	0.836
XGBoost	No tail/EVT	22	+0.002891	-0.005607	+0.008860	0.316	+0.00587	-0.01180	+0.01829	0.326
XGBoost	No liquidity	1	+0.000012	-0.000033	+0.000082	0.148	+0.00083	-0.00051	+0.00224	0.098
XGBoost	No liquidity	5	-0.000047	-0.000520	+0.000514	0.446	-0.00014	-0.00282	+0.00312	0.466
XGBoost	No liquidity	10	+0.001717	-0.001334	+0.004632	0.140	+0.00467	-0.00448	+0.01308	0.158
XGBoost	No liquidity	22	-0.003760	-0.008627	+0.002054	0.886	-0.00787	-0.01820	+0.00495	0.860
XGBoost	No breadth/dispersion	1	-0.000004	-0.000040	+0.000053	0.362	+0.00010	-0.00120	+0.00129	0.410
XGBoost	No breadth/dispersion	5	+0.000369	-0.000013	+0.000849	0.030	+0.00259	-0.00027	+0.00551	0.032
XGBoost	No breadth/dispersion	10	-0.003040	-0.005445	-0.001109	0.998	-0.00802	-0.01446	-0.00290	0.998
XGBoost	No breadth/dispersion	22	-0.002106	-0.007086	+0.001172	0.924	-0.00423	-0.01475	+0.00259	0.910
XGBoost	No market state	1	-0.000012	-0.000058	+0.000086	0.304	+0.00002	-0.00111	+0.00186	0.328
XGBoost	No market state	5	-0.000100	-0.000575	+0.000680	0.442	-0.00008	-0.00298	+0.00438	0.366
XGBoost	No market state	10	-0.000976	-0.003214	+0.001522	0.804	-0.00280	-0.00925	+0.00426	0.806
XGBoost	No market state	22	-0.005061	-0.010864	-0.001360	0.994	-0.01038	-0.02284	-0.00247	0.994
Dynamic ridge-logit	No GARCH volatility	1	+0.000005	-0.000041	+0.000087	0.320	-0.00117	-0.00243	+0.00002	0.974
Dynamic ridge-logit	No GARCH volatility	5	+0.000951	-0.000719	+0.003014	0.126	-0.00549	-0.01346	+0.00227	0.896
Dynamic ridge-logit	No GARCH volatility	10	-0.000719	-0.005205	+0.004219	0.582	-0.01806	-0.03471	-0.00245	0.990
Dynamic ridge-logit	No GARCH volatility	22	-0.012620	-0.019402	-0.006634	1.000	-0.05597	-0.08779	-0.03198	1.000
Dynamic ridge-logit	No volatility group	1	-0.000024	-0.000056	+0.000025	0.710	-0.00094	-0.00159	-0.00017	0.988
Dynamic ridge-logit	No volatility group	5	+0.000653	-0.000895	+0.002788	0.232	-0.01148	-0.02205	-0.00077	0.982
Dynamic ridge-logit	No volatility group	10	-0.000238	-0.004997	+0.005524	0.562	-0.02543	-0.04322	-0.00046	0.976
Dynamic ridge-logit	No volatility group	22	-0.015546	-0.024054	-0.007764	1.000	-0.08619	-0.11752	-0.04008	1.000
Dynamic ridge-logit	No tail/EVT	1	-0.000120	-0.000819	+0.000317	0.666	+0.00828	+0.00076	+0.01454	0.014
Dynamic ridge-logit	No tail/EVT	5	-0.005303	-0.014743	+0.004280	0.840	+0.02656	-0.01533	+0.05616	0.208
Dynamic ridge-logit	No tail/EVT	10	-0.043416	-0.073246	-0.019336	1.000	-0.00894	-0.14103	+0.07470	0.792
Dynamic ridge-logit	No tail/EVT	22	+0.011626	-0.040756	+0.043880	0.514	+0.35377	-0.19720	+0.58802	0.452
Dynamic ridge-logit	No liquidity	1	+0.000071	-0.000042	+0.000228	0.108	-0.00054	-0.00141	+0.00032	0.902
Dynamic ridge-logit	No liquidity	5	+0.000065	-0.002007	+0.002346	0.480	+0.00102	-0.00480	+0.00660	0.360

Model family	Ablation	Horizon	Full-ablation Δ Brier	Brier CI 2.5%	Brier CI 97.5%	P (full better): Brier	Full-ablation Δ Log Score	Log CI 2.5%	Log CI 97.5%	P (full better): Log
Dynamic ridge-logit	No liquidity	10	-0.001096	-0.006076	+0.004170	0.626	+0.00620	-0.01166	+0.02527	0.230
Dynamic ridge-logit	No liquidity	22	+0.010172	+0.003094	+0.018194	0.002	+0.03178	+0.00421	+0.06387	0.008
Dynamic ridge-logit	No breadth/dispersion	1	-0.000025	-0.000064	+0.000014	0.892	-0.00191	-0.00351	-0.00081	1.000
Dynamic ridge-logit	No breadth/dispersion	5	+0.000633	-0.001329	+0.002577	0.222	-0.00060	-0.00712	+0.00579	0.510
Dynamic ridge-logit	No breadth/dispersion	10	+0.005062	-0.000095	+0.011006	0.030	+0.00595	-0.00977	+0.02214	0.198
Dynamic ridge-logit	No breadth/dispersion	22	-0.004006	-0.012477	+0.002203	0.928	-0.04010	-0.07067	-0.01520	1.000
Dynamic ridge-logit	No market state	1	-0.000017	-0.000131	+0.000124	0.678	-0.00363	-0.00701	-0.00120	0.998
Dynamic ridge-logit	No market state	5	+0.001581	-0.001973	+0.005848	0.208	-0.00352	-0.01795	+0.01212	0.618
Dynamic ridge-logit	No market state	10	+0.003506	-0.007137	+0.014793	0.244	+0.03775	+0.00262	+0.08102	0.014
Dynamic ridge-logit	No market state	22	+0.031444	+0.014544	+0.049715	0.000	+0.15084	+0.08458	+0.25711	0.000

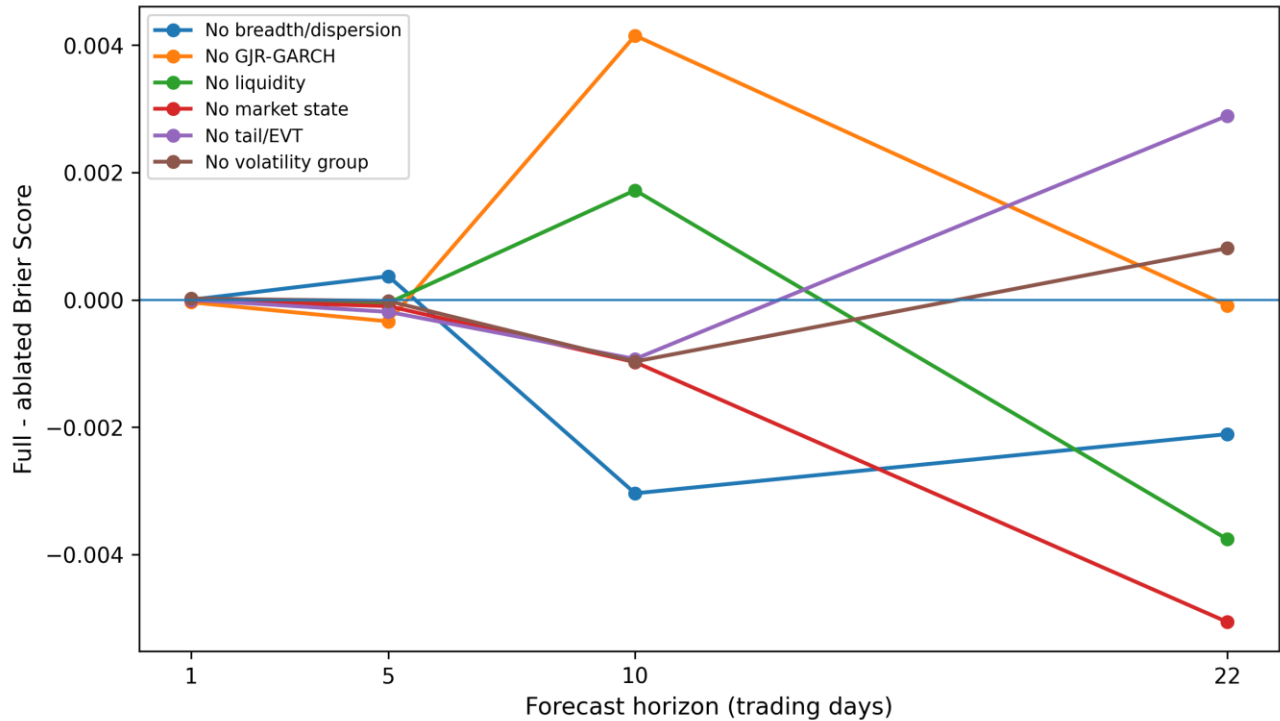
Note. Differences equal the Full specification minus the corresponding ablation. Negative Brier Score and Log Score differences therefore favour the Full specification because lower probability loss is better. Bootstrap inference uses 500 replications with 22-trading-day moving blocks. P (full better) is the proportion of bootstrap replications in which the Full specification records lower loss than the ablated model; values closer to 1 indicate stronger bootstrap evidence in favour of retaining that predictor component.

The component ablations show that the forecasting contribution of volatility information depends on both model and horizon. For XGBoost, removing the GJR-GARCH variables produces only small Brier Score differences at 1, 5, and 22 days, while the 10-day ablation performs better than the Full specification. Removing the broader volatility block instead worsens the 10-day Brier Score, although the corresponding 95% bootstrap interval includes zero. The dynamic ridge-logit displays a clearer long-horizon dependence on volatility information: at 22 days, the Full specification outperforms both the No GJR-GARCH and No volatility-group variants, with Brier differences of -0.01262 and -0.01555 and bootstrap intervals entirely below zero. The volatility evidence therefore supports H1 only partially because its incremental contribution varies across models and horizons.

The remaining ablations reinforce this model- and horizon-specific pattern. For XGBoost, removing breadth and dispersion at 10 days worsens the Brier Score by 0.00304 relative to the Full specification, while removing market-state information at 22 days worsens it by 0.00506; both bootstrap intervals exclude zero. Tail/EVT and liquidity contributions are less uniform, and some dynamic ridge-logit ablations change direction across horizons. These results show that no predictor block provides a universal improvement across the forecasting system. Instead, volatility, tail, liquidity, breadth/dispersion, and market-state information contribute selectively by forecasting model and horizon.

Horizon-specific changes in XGBoost Brier Score after removing each predictor component are summarised in Figure S3.

Figure S3. XGBoost full-minus-ablation Brier Score differences across forecast horizons



Note. The vertical axis reports the Full-model Brier Score minus the Brier Score of each ablated specification. Negative values indicate that retaining the component lowers Brier Score and therefore favours the Full model, whereas positive values indicate that the corresponding ablation performs better. The horizontal zero line indicates equal Brier Score performance.

The component effects vary substantially across forecast horizons. At 10 days, retaining breadth and dispersion information produces the clearest improvement in the Full model, whereas removing the GJR-GARCH variables improves performance at the same horizon. At 22 days, retaining market-state and liquidity information produces the largest negative point estimates, while removing the tail/EVT block slightly favours the ablated specification. These contrasting patterns show that the incremental forecasting value of individual predictor groups depends on the horizon rather than remaining uniformly beneficial across the XGBoost forecasts.

Crash-definition robustness first examines whether alternative tail definitions change event prevalence and identify materially different crash days. Table S11 reports target prevalence across the Train, Validation, and Test samples and then compares the overlap between the primary EVT and fixed-threshold definitions. Subsequent forecast comparisons assess whether these differences translate into changes in predictive performance.

Table S11. Crash-definition target prevalence and EVT-versus-fixed overlap**Panel A: Target prevalence by definition and sample split**

Definition	Split	Horizon	Valid targets	Events	Event rate (%)
EVT 1%	Train	0	5780	64	1.11
EVT 1%	Train	1	5652	63	1.11
EVT 1%	Train	5	5291	301	5.69
EVT 1%	Train	10	5039	548	10.88
EVT 1%	Train	22	4713	1049	22.26
EVT 1%	Validation	0	2933	31	1.06
EVT 1%	Validation	1	2889	31	1.07
EVT 1%	Validation	5	2738	143	5.22
EVT 1%	Validation	10	2598	273	10.51
EVT 1%	Validation	22	2359	526	22.30
EVT 1%	Test	0	2968	21	0.71
EVT 1%	Test	1	2960	21	0.71
EVT 1%	Test	5	2928	102	3.48
EVT 1%	Test	10	2888	202	6.99
EVT 1%	Test	22	2803	436	15.55
EVT 2.5%	Train	0	5780	141	2.44
EVT 2.5%	Train	1	5652	138	2.44
EVT 2.5%	Train	5	5315	628	11.82
EVT 2.5%	Train	10	5094	1114	21.87
EVT 2.5%	Train	22	4816	1935	40.18
EVT 2.5%	Validation	0	2933	73	2.49
EVT 2.5%	Validation	1	2889	73	2.53
EVT 2.5%	Validation	5	2738	342	12.49
EVT 2.5%	Validation	10	2605	645	24.76
EVT 2.5%	Validation	22	2402	1152	47.96
EVT 2.5%	Test	0	2968	66	2.22
EVT 2.5%	Test	1	2960	66	2.23
EVT 2.5%	Test	5	2928	304	10.38
EVT 2.5%	Test	10	2890	581	20.10
EVT 2.5%	Test	22	2832	1139	40.22
EVT 5%	Train	0	5780	304	5.26
EVT 5%	Train	1	5652	296	5.24
EVT 5%	Train	5	5348	1251	23.39
EVT 5%	Train	10	5174	2095	40.49
EVT 5%	Train	22	5007	3339	66.69
EVT 5%	Validation	0	2933	150	5.11
EVT 5%	Validation	1	2889	150	5.19
EVT 5%	Validation	5	2740	683	24.93
EVT 5%	Validation	10	2621	1156	44.11
EVT 5%	Validation	22	2476	1856	74.96
EVT 5%	Test	0	2968	128	4.31
EVT 5%	Test	1	2960	128	4.32
EVT 5%	Test	5	2931	553	18.87
EVT 5%	Test	10	2899	1010	34.84
EVT 5%	Test	22	2853	1749	61.30
Fixed 2.5%	Train	0	5780	147	2.54
Fixed 2.5%	Train	1	5652	143	2.53
Fixed 2.5%	Train	5	5321	578	10.86
Fixed 2.5%	Train	10	5107	987	19.33
Fixed 2.5%	Train	22	4855	1730	35.63
Fixed 2.5%	Validation	0	2933	81	2.76
Fixed 2.5%	Validation	1	2889	81	2.80

Definition	Split	Horizon	Valid targets	Events	Event rate (%)
Fixed 2.5%	Validation	5	2739	346	12.63
Fixed 2.5%	Validation	10	2611	619	23.71
Fixed 2.5%	Validation	22	2435	1063	43.66
Fixed 2.5%	Test	0	2968	91	3.07
Fixed 2.5%	Test	1	2960	91	3.07
Fixed 2.5%	Test	5	2931	377	12.86
Fixed 2.5%	Test	10	2899	663	22.87
Fixed 2.5%	Test	22	2848	1147	40.27

Note. Horizon 0 reports the contemporaneous same-day crash label, whereas horizons 1, 5, 10, and 22 indicate whether at least one crash occurs within the corresponding forward trading-day window. The longer-horizon event rates are therefore cumulative target frequencies and are not expected to equal the nominal one-day tail probabilities. Valid targets report observations for which the corresponding horizon-specific outcome is defined. EVT 2.5% is the primary crash definition; EVT 1% and EVT 5% provide tail-probability robustness, while Fixed 2.5% provides the unconditional-threshold comparison

Same-day prevalence follows the intended ordering across the three EVT definitions. In the Test sample, the EVT 1%, 2.5%, and 5% definitions identify crash rates of 0.71%, 2.22%, and 4.31%, respectively, while the fixed 2.5% rule identifies a higher rate of 3.07%. Event prevalence rises mechanically with the forecast horizon because the target records whether a crash occurs anywhere within the forward window. The differences between the EVT 2.5% and fixed 2.5% rates already indicate that volatility adjustment changes the resulting crash classification, although the degree of event-by-event overlap is examined directly in Panel B.

Table S11. Crash-definition prevalence and EVT-versus-fixed overlap
Panel B. Same-day overlap between EVT 2.5% and fixed 2.5% crash definitions

Split	Sector	Common days	EVT events	Fixed events	Common events	Jaccard	Agreement rate
Train	Banking	994	20	25	14	0.452	0.983
Train	Commercial and services	983	21	25	9	0.243	0.972
Train	Energy and Petroleum	973	25	25	20	0.667	0.990
Train	Insurance	995	29	25	19	0.543	0.984
Train	Investment	994	32	25	15	0.357	0.973
Train	Manufacturing and Allied	841	14	22	12	0.500	0.986
Validation	Banking	496	8	6	5	0.556	0.992
Validation	Commercial and services	483	10	4	2	0.167	0.979
Validation	Energy and Petroleum	495	13	14	11	0.688	0.990
Validation	Insurance	496	16	14	10	0.500	0.980
Validation	Investment	489	14	19	10	0.435	0.973
Validation	Manufacturing and Allied	474	12	24	12	0.500	0.975
Test	Banking	495	15	8	8	0.533	0.986
Test	Commercial and services	494	10	15	7	0.389	0.978
Test	Energy and Petroleum	495	6	8	4	0.400	0.988
Test	Insurance	495	10	9	5	0.357	0.982
Test	Investment	495	10	22	7	0.280	0.964
Test	Manufacturing and Allied	494	15	29	15	0.517	0.972

Note. Jaccard overlap equals the number of crash events identified by both definitions divided by the number identified by either definition. Agreement rate is the proportion of common sector-days assigned the same crash/non-crash classification by both definitions. Because crashes are relatively infrequent, agreement rates are dominated by jointly classified non-crash days; Jaccard overlap therefore provides the more informative measure of overlap between the positive crash-event sets.

The EVT and fixed-threshold definitions identify materially different sets of same-day crash events despite using the same nominal 2.5% tail probability. In the Test sample, the EVT rule identifies 66 crashes, and the fixed-threshold rule identifies 91, with only 46 events in common. Sector-level Jaccard overlap ranges from 0.280 for Investment to 0.533 for Banking, averaging about 0.41 across the six sectors. This cross-sectional

divergence reflects the varying volatility regimes of the underlying industries; the low overlap in the Investment sector highlights its vulnerability to volatility clustering, which causes the unconditional fixed threshold to overcount consecutive event sequences during prolonged distress. The high overall agreement rates mainly reflect the large number of days both rules classify as non-crashes and therefore do not imply close agreement on the crash events themselves. These results show that volatility conditioning changes which returns are classified as crashes rather than merely relabeling essentially the same extreme-return observations. Ultimately, this confirms that dynamic conditioning strips out cyclical regime noise, isolating structural, idiosyncratic innovations that are fundamentally invisible to a static threshold framework.

Table S12 reports pooled Test performance under the 1%, 2.5%, and 5% EVT definitions and the fixed 2.5% return-threshold definition. The comparison includes fixed and expanding historical benchmarks, XGBoost, and the dynamic ridge-logit model across the four forecast horizons. Because event prevalence differs across crash definitions, the study interprets raw probability losses within each definition rather than comparing them directly across definitions.

Table S12. Crash-definition forecast robustness: pooled Test performance

Definition	Model family	Horizon	N	Events	Event rate (%)	Brier	Log Score	PR-AUC	ROC-AUC	Calibration slope
EVT 1%	Fixed Historical	1	2960	21	0.71	0.0071	0.0432	0.0074	0.5195	0.1190
EVT 1%	Fixed Historical	5	2928	102	3.48	0.0341	0.1564	0.0360	0.5173	0.0679
EVT 1%	Fixed Historical	10	2888	202	6.99	0.0668	0.2633	0.0722	0.5160	0.0414
EVT 1%	Fixed Historical	22	2803	436	15.55	0.1371	0.4498	0.1608	0.5173	0.0361
EVT 1%	Expanding Historical	1	2960	21	0.71	0.0071	0.0427	0.0076	0.5227	0.7573
EVT 1%	Expanding Historical	5	2928	102	3.48	0.0339	0.1540	0.0341	0.5141	0.6430
EVT 1%	Expanding Historical	10	2888	202	6.99	0.0660	0.2585	0.0675	0.5125	0.6091
EVT 1%	Expanding Historical	22	2803	436	15.55	0.1339	0.4391	0.1517	0.5197	0.6656
EVT 1%	XGBoost	1	2960	21	0.71	0.0070	0.0413	0.0135	0.7140	1.6605
EVT 1%	XGBoost	5	2928	102	3.48	0.0336	0.1509	0.0486	0.6485	2.4380
EVT 1%	XGBoost	10	2888	202	6.99	0.0660	0.2590	0.0783	0.5698	2.6165
EVT 1%	XGBoost	22	2803	436	15.55	0.1359	0.4465	0.1555	0.5000	1.2163
EVT 1%	Dynamic Logit	1	2960	21	0.71	0.0126	0.0807	0.0121	0.5852	0.0476
EVT 1%	Dynamic Logit	5	2928	102	3.48	0.0656	0.2634	0.0525	0.5613	0.0328
EVT 1%	Dynamic Logit	10	2888	202	6.99	0.1628	0.5826	0.0830	0.5441	0.0046
EVT 1%	Dynamic Logit	22	2803	436	15.55	0.2252	0.7876	0.1598	0.5350	-0.0227
EVT 2.5%	Fixed Historical	1	2960	66	2.23	0.0219	0.1085	0.0203	0.4429	-0.8604
EVT 2.5%	Fixed Historical	5	2928	304	10.38	0.0945	0.3406	0.0963	0.4506	-0.7136
EVT 2.5%	Fixed Historical	10	2890	581	20.10	0.1652	0.5157	0.1873	0.4452	-0.7485
EVT 2.5%	Fixed Historical	22	2832	1139	40.22	0.2499	0.6939	0.3915	0.4608	-0.4557
EVT 2.5%	Expanding Historical	1	2960	66	2.23	0.0219	0.1080	0.0200	0.4244	-1.1838
EVT 2.5%	Expanding Historical	5	2928	304	10.38	0.0941	0.3385	0.0919	0.4288	-0.9885
EVT 2.5%	Expanding Historical	10	2890	581	20.10	0.1637	0.5110	0.1797	0.4268	-0.9422
EVT 2.5%	Expanding Historical	22	2832	1139	40.22	0.2462	0.6859	0.3859	0.4431	-0.4999
EVT 2.5%	XGBoost	1	2960	66	2.23	0.0218	0.1064	0.0480	0.6251	0.7563
EVT 2.5%	XGBoost	5	2928	304	10.38	0.0919	0.3269	0.1562	0.6190	1.1581
EVT 2.5%	XGBoost	10	2890	581	20.10	0.1618	0.5052	0.2403	0.5572	1.8694
EVT 2.5%	XGBoost	22	2832	1139	40.22	0.2370	0.6667	0.4826	0.5719	0.9193
EVT 2.5%	Dynamic Logit	1	2960	66	2.23	0.0219	0.1201	0.0420	0.6381	0.1479
EVT 2.5%	Dynamic Logit	5	2928	304	10.38	0.0979	0.4082	0.1639	0.6283	0.1225
EVT 2.5%	Dynamic Logit	10	2890	581	20.10	0.1815	0.6868	0.2681	0.6013	0.0824
EVT 2.5%	Dynamic Logit	22	2832	1139	40.22	0.3073	1.1943	0.4737	0.5902	0.0522
EVT 5%	Fixed Historical	1	2960	128	4.32	0.0416	0.1804	0.0398	0.4507	-1.4850

Definition	Model family	Horizon	N	Events	Event rate (%)	Brier	Log Score	PR-AUC	ROC-AUC	Calibration slope
EVT 5%	Fixed Historical	5	2931	553	18.87	0.1574	0.4965	0.1757	0.4494	-1.3266
EVT 5%	Fixed Historical	10	2899	1010	34.84	0.2367	0.6666	0.3290	0.4439	-1.2936
EVT 5%	Fixed Historical	22	2853	1749	61.30	0.2500	0.6974	0.6036	0.4478	-0.9886
EVT 5%	Expanding Historical	1	2960	128	4.32	0.0415	0.1793	0.0405	0.4514	-1.2360
EVT 5%	Expanding Historical	5	2931	553	18.87	0.1556	0.4916	0.1780	0.4587	-0.8788
EVT 5%	Expanding Historical	10	2899	1010	34.84	0.2322	0.6573	0.3334	0.4556	-0.6637
EVT 5%	Expanding Historical	22	2853	1749	61.30	0.2429	0.6801	0.6243	0.4708	0.2267
EVT 5%	XGBoost	1	2960	128	4.32	0.0410	0.1747	0.0880	0.6336	1.1701
EVT 5%	XGBoost	5	2931	553	18.87	0.1555	0.4912	0.2045	0.5418	1.2593
EVT 5%	XGBoost	10	2899	1010	34.84	0.2349	0.6628	0.3860	0.5382	7.2785
EVT 5%	XGBoost	22	2853	1749	61.30	0.2419	0.6783	0.7066	0.6226	1.8546
EVT 5%	Dynamic Logit	1	2960	128	4.32	0.0425	0.2145	0.0666	0.6207	0.0903
EVT 5%	Dynamic Logit	5	2931	553	18.87	0.1747	0.6118	0.2195	0.5763	0.1507
EVT 5%	Dynamic Logit	10	2899	1010	34.84	0.2829	0.9123	0.3825	0.5685	0.1023
EVT 5%	Dynamic Logit	22	2853	1749	61.30	0.3909	1.2817	0.6543	0.5678	0.0759
Fixed 2.5%	Fixed Historical	1	2960	91	3.07	0.0298	0.1377	0.0419	0.5750	6.4346
Fixed 2.5%	Fixed Historical	5	2931	377	12.86	0.1120	0.3834	0.1825	0.5900	7.2903
Fixed 2.5%	Fixed Historical	10	2899	663	22.87	0.1760	0.5365	0.3206	0.5964	10.0000
Fixed 2.5%	Fixed Historical	22	2848	1147	40.27	0.2406	0.6743	0.5047	0.5841	8.3038
Fixed 2.5%	Expanding Historical	1	2960	91	3.07	0.0297	0.1356	0.0404	0.5892	1.3329
Fixed 2.5%	Expanding Historical	5	2931	377	12.86	0.1097	0.3737	0.1730	0.6068	1.5465
Fixed 2.5%	Expanding Historical	10	2899	663	22.87	0.1699	0.5207	0.3027	0.6083	1.5358
Fixed 2.5%	Expanding Historical	22	2848	1147	40.27	0.2299	0.6526	0.4997	0.6114	1.3840
Fixed 2.5%	XGBoost	1	2960	91	3.07	0.0296	0.1341	0.0735	0.7208	1.1696
Fixed 2.5%	XGBoost	5	2931	377	12.86	0.1103	0.3783	0.2244	0.6447	1.0203
Fixed 2.5%	XGBoost	10	2899	663	22.87	0.1690	0.5185	0.4022	0.6202	1.7330
Fixed 2.5%	XGBoost	22	2848	1147	40.27	0.2300	0.6536	0.5680	0.6302	1.0892
Fixed 2.5%	Dynamic Logit	1	2960	91	3.07	0.0337	0.1548	0.0870	0.6085	0.3896
Fixed 2.5%	Dynamic Logit	5	2931	377	12.86	0.1910	0.8936	0.1423	0.5144	0.0174
Fixed 2.5%	Dynamic Logit	10	2899	663	22.87	0.3692	1.1587	0.2606	0.4879	0.0373
Fixed 2.5%	Dynamic Logit	22	2848	1147	40.27	0.3476	1.1012	0.4532	0.5336	0.0850

Note. Crash definitions produce different event prevalences, so raw Brier Score and Log Score values should not be compared directly across definitions. Within each definition, lower Brier Score and Log Score indicate better probability forecasts. Cross-definition comparisons therefore use forecast skill relative to each definition's own historical benchmark.

Under the primary EVT 2.5% definition, XGBoost records a lower Brier Score than the expanding historical benchmark at all four horizons, with the largest reduction occurring at 22 days. The alternative EVT definitions provide a less uniform pattern: XGBoost improves on the corresponding historical benchmark at several horizons under both the 1% and 5% definitions, but not consistently across all four horizons. The fixed 2.5% definition also produces mixed relative performance, with only small Brier differences from its historical benchmark. These results show that forecast ability depends partly on the crash definition and tail probability, while the primary EVT specification provides the most consistent XGBoost improvement across horizons.

Table S12. Panel B. EVT 2.5% versus fixed 2.5% definition-specific Brier-skill comparison

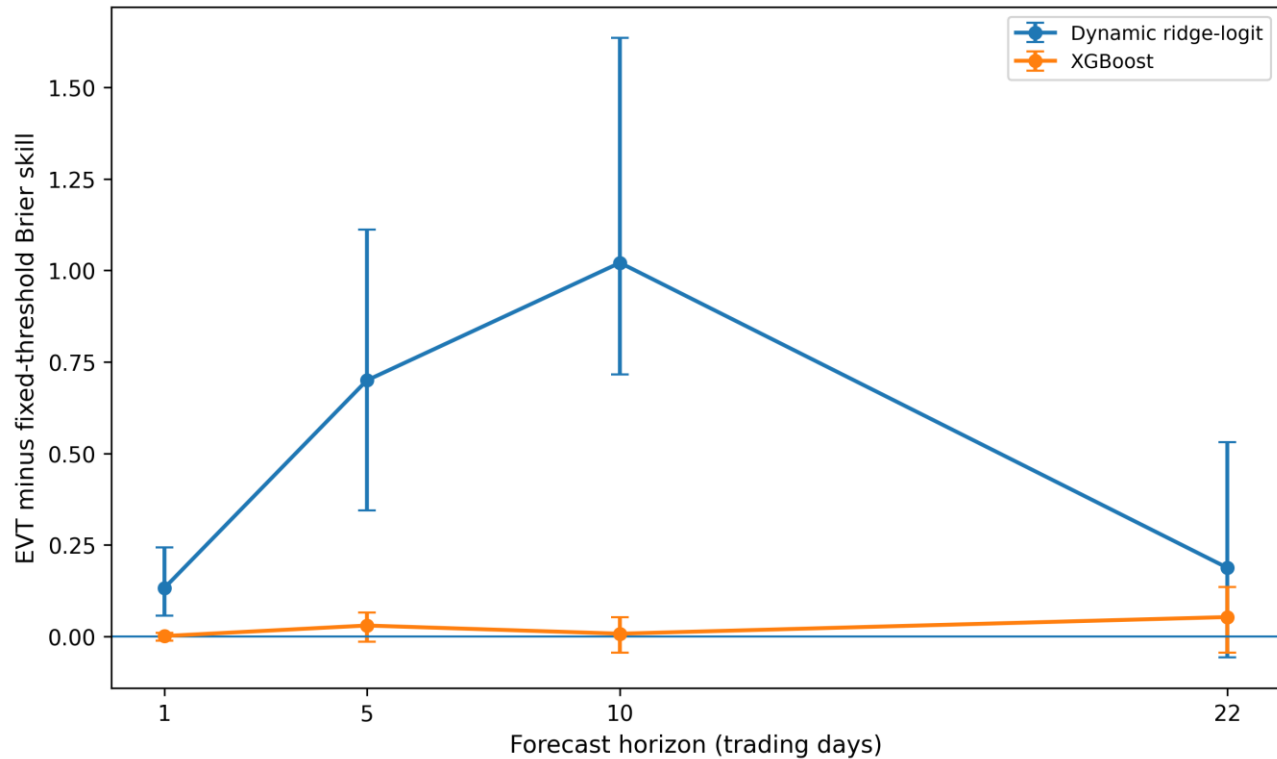
Model	Horizon	EVT Brier skill	Fixed Brier skill	EVT-fixed difference	CI 2.5%	CI 97.5%	P (EVT better)
XGBoost	1	0.003881	0.002822	0.001059	-0.011758	0.009285	0.416
XGBoost	5	0.023036	-0.006805	0.029842	-0.014491	0.065358	0.924
XGBoost	10	0.011510	0.003995	0.007515	-0.044656	0.052803	0.610

Model	Horizon	EVT Brier skill	Fixed Brier skill	EVT-fixed difference	CI 2.5%	CI 97.5%	P (EVT better)
XGBoost	22	0.032859	-0.020067	0.052926	-0.044222	0.135358	0.834
Dynamic ridge-logit	1	-0.003103	-0.135183	0.132079	0.056507	0.243688	1.000
Dynamic ridge-logit	5	-0.041367	-0.740953	0.699586	0.344866	1.111464	1.000
Dynamic ridge-logit	10	-0.109640	-1.130777	1.021137	0.715601	1.635149	1.000
Dynamic ridge-logit	22	-0.282620	-0.470056	0.187436	-0.056817	0.530579	0.928

Note. Each Brier-skill measure is calculated relative to the historical benchmark for the same crash definition. The difference equals EVT Brier skill minus fixed-threshold Brier skill, so positive values favour EVT. Bootstrap inference uses 500 replications with 22-trading-day moving blocks. P (EVT better) is the proportion of bootstrap replications in which the EVT definition produces the higher definition-specific Brier skill.

Definition-specific Brier skill provides a common relative-performance scale for comparing the EVT 2.5% and fixed 2.5% crash definitions, as shown in Figure S4.

Figure S4. EVT 2.5% versus fixed 2.5% definition-specific Brier-skill differences.



Note. Each point reports EVT Brier skill minus fixed-threshold Brier skill, with each skill measure calculated relative to its own historical benchmark. Positive values therefore favour the EVT 2.5% definition, whereas negative values favour the fixed 2.5% definition. Error bars show 95% confidence intervals from the 22-trading-day moving-block bootstrap.

The XGBoost point estimates favour the EVT definition at all four horizons, but the corresponding 95% bootstrap intervals include zero, so the evidence is directional rather than statistically decisive. The dynamic ridge-logit produces substantially larger positive differences at several horizons, indicating stronger relative performance under the EVT definition, although these gains should be interpreted against each definition's own historical benchmark rather than as evidence of uniformly strong absolute forecast skill. The results therefore provide partial support for H2: volatility-adjusted EVT event identification generally improves relative forecast skill, but the advantage is model- and horizon-dependent rather than universal.

SI G. Sector/horizon calibration and discrimination results

Pooled volatility-ablation results may conceal differences across sectors and forecast horizons. Table S13 therefore reports sector-specific Brier Scores for the Full, No GJR-GARCH, and No volatility-group specifications for XGBoost and the dynamic ridge-logit benchmark, together with the Full-model PR-AUC.

Table S13. Sector-by-horizon volatility-ablation heterogeneity

Model family	Sector	Horizon	Full Brier	Full PR-AUC	No GJR-GARCH Brier	No volatility-group Brier	GJR-GARCH increment	Volatility-group increment
XGBoost	Banking	1	0.02961	0.04728	0.02967	0.02953	0.00006	-0.00007
XGBoost	Commercial and services	1	0.01984	0.12330	0.01993	0.01990	0.00009	0.00007
XGBoost	Energy and Petroleum	1	0.01196	0.03027	0.01193	0.01195	-0.00003	-0.00002
XGBoost	Insurance	1	0.01971	0.08135	0.01973	0.01961	0.00002	-0.00010
XGBoost	Investment	1	0.01979	0.04063	0.01980	0.01975	0.00001	-0.00004
XGBoost	Manufacturing and Allied	1	0.02973	0.05593	0.02980	0.02977	0.00007	0.00004
XGBoost	Banking	5	0.11498	0.14990	0.11620	0.11631	0.00122	0.00133
XGBoost	Commercial and services	5	0.08527	0.18733	0.08603	0.08469	0.00076	-0.00058
XGBoost	Energy and Petroleum	5	0.05611	0.13436	0.05624	0.05643	0.00012	0.00032
XGBoost	Insurance	5	0.08922	0.27369	0.09018	0.08964	0.00096	0.00042
XGBoost	Investment	5	0.08513	0.15854	0.08473	0.08495	-0.00041	-0.00019
XGBoost	Manufacturing and Allied	5	0.12115	0.18100	0.12053	0.12002	-0.00063	-0.00113
XGBoost	Banking	10	0.18645	0.25517	0.18238	0.18516	-0.00407	-0.00128
XGBoost	Commercial and services	10	0.14840	0.25026	0.14832	0.14970	-0.00009	0.00130
XGBoost	Energy and Petroleum	10	0.11998	0.16591	0.11014	0.12376	-0.00983	0.00378
XGBoost	Insurance	10	0.15849	0.24278	0.15744	0.16018	-0.00105	0.00170
XGBoost	Investment	10	0.15590	0.21166	0.14805	0.15706	-0.00785	0.00116
XGBoost	Manufacturing and Allied	10	0.20208	0.29984	0.20027	0.20129	-0.00181	-0.00078
XGBoost	Banking	22	0.26302	0.41778	0.25760	0.24582	-0.00542	-0.01720
XGBoost	Commercial and services	22	0.23779	0.38655	0.24787	0.25342	0.01008	0.01563
XGBoost	Energy and Petroleum	22	0.20587	0.40368	0.19523	0.20426	-0.01064	-0.00161
XGBoost	Insurance	22	0.24292	0.56025	0.24197	0.24212	-0.00095	-0.00080
XGBoost	Investment	22	0.22420	0.63287	0.21986	0.22014	-0.00434	-0.00406
XGBoost	Manufacturing and Allied	22	0.24849	0.55906	0.26032	0.25676	0.01183	0.00827
Dynamic ridge-logit	Banking	1	0.02980	0.04203	0.02983	0.02982	0.00003	0.00002
Dynamic ridge-logit	Commercial and services	1	0.02049	0.06514	0.02025	0.02048	-0.00024	-0.00001
Dynamic ridge-logit	Energy and Petroleum	1	0.01203	0.02724	0.01202	0.01203	-0.00001	-0.00001
Dynamic ridge-logit	Insurance	1	0.01981	0.06531	0.01985	0.01984	0.00003	0.00002
Dynamic ridge-logit	Investment	1	0.01952	0.14110	0.01965	0.01964	0.00013	0.00012
Dynamic ridge-logit	Manufacturing and Allied	1	0.02990	0.08717	0.02992	0.02990	0.00003	0.00000
Dynamic ridge-logit	Banking	5	0.10806	0.24294	0.10932	0.10983	0.00126	0.00177
Dynamic ridge-logit	Commercial and services	5	0.11662	0.20227	0.10848	0.10836	-0.00814	-0.00826
Dynamic ridge-logit	Energy and Petroleum	5	0.05765	0.13428	0.05704	0.05678	-0.00061	-0.00087
Dynamic ridge-logit	Insurance	5	0.09431	0.20640	0.09520	0.09541	0.00089	0.00110
Dynamic ridge-logit	Investment	5	0.08668	0.19735	0.08809	0.08843	0.00142	0.00175
Dynamic ridge-logit	Manufacturing and Allied	5	0.12468	0.30645	0.12408	0.12533	-0.00060	0.00065
Dynamic ridge-logit	Banking	10	0.19377	0.38662	0.19364	0.18827	-0.00013	-0.00550
Dynamic ridge-logit	Commercial and services	10	0.22385	0.24089	0.21991	0.21966	-0.00394	-0.00420
Dynamic ridge-logit	Energy and Petroleum	10	0.11750	0.20593	0.11319	0.11157	-0.00431	-0.00593
Dynamic ridge-logit	Insurance	10	0.17130	0.25607	0.17592	0.17681	0.00462	0.00551
Dynamic ridge-logit	Investment	10	0.15448	0.33055	0.16236	0.16362	0.00788	0.00914
Dynamic ridge-logit	Manufacturing and Allied	10	0.23015	0.39255	0.23047	0.23301	0.00032	0.00286
Dynamic ridge-logit	Banking	22	0.21662	0.68943	0.22475	0.23182	0.00813	0.01520
Dynamic ridge-logit	Commercial and services	22	0.39693	0.45441	0.39996	0.39594	0.00302	-0.00099
Dynamic ridge-logit	Energy and Petroleum	22	0.19717	0.48274	0.19981	0.20326	0.00264	0.00609
Dynamic ridge-logit	Insurance	22	0.34603	0.55926	0.36240	0.36494	0.01637	0.01891
Dynamic ridge-logit	Investment	22	0.28776	0.62439	0.32616	0.32808	0.03840	0.04032
Dynamic ridge-logit	Manufacturing and Allied	22	0.40271	0.65312	0.40977	0.41640	0.00706	0.01369

Note. The GARCH-volatility increment equals the No GARCH-volatility Brier Score minus the Full-model Brier Score and isolates the contribution of GJR-GARCH conditional volatility. The volatility-group increment equals the No volatility-group Brier Score minus the Full-model Brier Score and assesses the broader block comprising GJR-GARCH conditional volatility and the Parkinson, Garman-Klass, and Rogers-Satchell range-based volatility measures. Positive increments favour the Full specification.

The sector-level results show substantial variation in the predictive contribution of volatility information across both models and horizons. For XGBoost, the isolated GJR-GARCH contribution is small at 1 and 5 days and becomes negative across all six sectors at 10 days, whereas the broader volatility group still improves the Full model in four of the six sectors at that horizon. At 22 days, both GJR-GARCH and broader volatility contributions remain mixed, with positive effects for Commercial and Services and Manufacturing and Allied but negative effects for several other sectors.

The dynamic ridge-logit displays a different pattern. Its volatility contributions remain mixed at shorter horizons, but at 22 days the GJR-GARCH increment is positive in all six sectors, and the broader volatility-group increment is positive in five of six sectors, with the largest gains occurring for Investment and Insurance. These differences show that volatility information does not provide a uniform market-wide forecasting benefit. Instead, its incremental value depends jointly on the forecasting model, sector, and horizon, reinforcing the partial support for H1 and the sector-horizon heterogeneity underlying H5.

XGBoost gain importance provides a descriptive audit of how strongly the original predictors contribute to split improvements in the fitted primary model. Table S14 reports the complete gain-importance distribution after aggregating derived inputs to their original feature names at each forecast horizon.

Table S14. Aggregated XGBoost gain importance by forecast horizon

Horizon	Original feature	Gain	Gain share (%)
1	EVT_u	425.28971	5.629
1	EVT_pu	360.24769	4.768
1	Sector Return_Model	343.88426	4.552
1	EVT_nhist	276.02033	3.654
1	StdInnovation	261.21224	3.458
1	GARCH_Sigma	238.15436	3.152
1	Market Return Dispersion	233.60001	3.092
1	GARCH_Mu	230.80622	3.055
1	EVT_scale	227.96101	3.017
1	Market Zero Return Share	219.75722	2.909
1	Market CapKES_Total	208.35313	2.758
1	AmihudILLIQ_Median	202.80582	2.684
1	NStocks	191.50383	2.535
1	NStocks Return	183.71400	2.432
1	EVT_shape	179.62678	2.378
1	Market Breadth Negative	173.62704	2.298
1	Sector Day Eligible	170.49518	2.257
1	EVT_nexc	168.95956	2.236
1	Zero Volume Share	161.68398	2.140
1	Sector Return_MCW	157.99592	2.091
1	Market Zero Volume Share	151.57190	2.006
1	NStocks Observed Market	145.62105	1.928
1	NSectors With Return	144.12944	1.908
1	NStocksObserved	142.38331	1.885
1	Sector Return_EW	141.89057	1.878
1	Market Turnover_MCW	138.09644	1.828
1	Sector Observed	133.40952	1.766
1	Return Dispersion	132.38119	1.752
1	Turnover_MCW	129.79408	1.718
1	NCorporate Actions	125.86477	1.666
1	NStocks Return Market	124.96569	1.654
1	Market Cap KES	116.56666	1.543

Horizon	Original feature	Gain	Gain share (%)
1	NSectors Down	116.56446	1.543
1	RogersSatchellVol	109.60896	1.451
1	Breadth Negative	106.86491	1.415
1	Market AmihudILLIQ Median	103.70742	1.373
1	GarmanKlassVol	103.20468	1.366
1	EVENT HISTORY	100.10159	1.325
1	ParkinsonVol	99.72732	1.320
1	Eligible Day Coverage	88.05088	1.166
1	Sector Return Available	86.66854	1.147
1	NSectors Observed	86.36574	1.143
1	Return Coverage	84.83951	1.123
1	Constituent Return Share	81.20922	1.075
1	HHI	61.07657	0.808
1	Zero Return Share	53.44221	0.707
1	Cross Sector Return Dispersion	16.98242	0.225
1	Market Return MCW	13.95603	0.185
5	EVT nhist	514.77874	5.917
5	EVT u	496.28345	5.704
5	EVT pu	462.67815	5.318
5	Sector Return Model	417.77556	4.802
5	GARCH Mu	404.81428	4.653
5	GARCH Sigma	333.26235	3.830
5	EVT scale	321.27398	3.692
5	StdInnovation	286.25347	3.290
5	NStocks	279.84802	3.216
5	Market CapKES Total	230.52582	2.650
5	Sector Day Eligible	225.46352	2.591
5	Market AmihudILLIQ Median	224.08818	2.576
5	EVT shape	223.94207	2.574
5	Return Coverage	223.02844	2.563
5	Market Breadth Negative	208.82946	2.400
5	NSectors With Return	197.16159	2.266
5	EVT nexc	186.18166	2.140
5	Eligible Day Coverage	186.03680	2.138
5	Turnover MCW	177.03156	2.035
5	Market Turnover MCW	176.41147	2.028
5	Breadth Negative	167.10941	1.921
5	NSectors Down	166.62556	1.915
5	NCorporateActions	164.93227	1.896
5	Market Zero Return Share	160.72117	1.847
5	NStocksObserved	155.10807	1.783
5	AmihudILLIQ Median	148.61861	1.708
5	NStocks Observed Market	144.44262	1.660
5	Market Zero Volume Share	142.30565	1.636
5	NSectors Observed	142.19583	1.634
5	Zero Volume Share	127.25562	1.463
5	Market Return Dispersion	126.85561	1.458
5	Return Dispersion	126.73257	1.457
5	Constituent Return Share	112.90599	1.298
5	Sector Return Available	110.18426	1.266
5	RogersSatchellVol	109.17781	1.255
5	Sector Observed	107.51805	1.236
5	NStocks Return	105.21785	1.209
5	Sector Return MCW	82.19858	0.945
5	Sector Return EW	81.32257	0.935
5	Market CapKES	78.28810	0.900

Horizon	Original feature	Gain	Gain share (%)
5	GarmanKlassVol	75.58688	0.869
5	NStocks Return Market	59.34398	0.682
5	ParkinsonVol	51.48850	0.592
5	EVENT HISTORY	49.63202	0.570
5	Zero Return Share	42.69670	0.491
5	Cross Sector Return Dispersion	34.39904	0.395
5	HHI	26.72397	0.307
5	Market Return MCW	25.47044	0.293
10	Sector Return Model	778.41502	6.829
10	EVT pu	771.30260	6.766
10	GARCH Sigma	734.38026	6.442
10	EVT u	721.67269	6.331
10	EVT nhist	643.41360	5.644
10	GARCH Mu	603.60558	5.295
10	NStocks	538.30082	4.722
10	StdInnovation	508.67526	4.462
10	NSectors With Return	455.35983	3.995
10	EVT scale	375.52182	3.294
10	Market Zero Volume Share	372.76113	3.270
10	EVT shape	355.03470	3.115
10	Market Breadth Negative	336.31976	2.950
10	EVT nexc	332.51941	2.917
10	NSectors Down	285.89079	2.508
10	Market AmihudILLIQ Median	272.20700	2.388
10	Market Turnover MCW	257.99733	2.263
10	NStocks Observed Market	256.48868	2.250
10	Sector Day Eligible	254.21802	2.230
10	Return Coverage	237.61359	2.084
10	NSectors Observed	223.16059	1.958
10	Market Zero Return Share	218.86843	1.920
10	NStocksObserved	176.46383	1.548
10	Sector Observed	139.45286	1.223
10	Market Return Dispersion	137.56713	1.207
10	Eligible Day Coverage	133.24503	1.169
10	Sector Return Available	130.31691	1.143
10	NStocks Return	127.17417	1.116
10	Zero Volume Share	124.71831	1.094
10	Garman KlassVol	114.08720	1.001
10	Breadth Negative	106.51744	0.934
10	Turnover MCW	100.87366	0.885
10	Market CapKES Total	100.62527	0.883
10	NStocks Return Market	97.45509	0.855
10	NCorporateActions	76.45030	0.671
10	EVENT HISTORY	74.89738	0.657
10	AmihudILLIQ Median	51.44396	0.451
10	Market Return MCW	48.98605	0.430
10	RogersSatchellVol	25.69174	0.225
10	Cross Sector Return Dispersion	24.68399	0.217
10	Market CapKES	22.25762	0.195
10	Constituent Return Share	20.79241	0.182
10	Return Dispersion	12.16612	0.107
10	Zero Return Share	11.68915	0.103
10	Sector Return MCW	7.82150	0.069
22	EVT pu	899.77199	8.741
22	EVT u	823.20772	7.997
22	GARCH Sigma	664.65390	6.457

Horizon	Original feature	Gain	Gain share (%)
22	GARCH Mu	589.41902	5.726
22	EVT_nhist	552.57016	5.368
22	StdInnovation	507.28227	4.928
22	EVT_shape	447.06168	4.343
22	NSectors With Return	430.76084	4.185
22	Market Return Dispersion	408.66544	3.970
22	EVT_scale	401.75391	3.903
22	NSectors Observed	365.99365	3.555
22	Sector Return Model	327.37718	3.180
22	NStocks	315.82895	3.068
22	NSectors Down	265.76677	2.582
22	EVT_nexc	244.40395	2.374
22	Return Coverage	244.16182	2.372
22	Market Turnover MCW	242.26770	2.353
22	Market CapKES Total	236.46341	2.297
22	Market Breadth Negative	219.79070	2.135
22	Garman KlassVol	203.45655	1.976
22	Sector Observed	187.50802	1.822
22	Market Zero Volume Share	180.16484	1.750
22	Market Zero Return Share	177.05478	1.720
22	NStocks Return Market	165.02052	1.603
22	Eligible Day Coverage	152.23195	1.479
22	Market AmihudILLIQ Median	140.28313	1.363
22	Cross Sector Return Dispersion	119.03436	1.156
22	NStocks Observed Market	117.95501	1.146
22	Breadth Negative	96.38882	0.936
22	NStocks Return	90.28959	0.877
22	Parkinson Vol	69.79621	0.678
22	EVENT HISTORY	68.12782	0.662
22	Sector Day Eligible	48.18069	0.468
22	AmihudILLIQ Median	41.37933	0.402
22	NStocksObserved	40.78370	0.396
22	Constituent Return Share	35.25282	0.342
22	RogersSatchellVol	32.37572	0.315
22	NCorporateActions	30.47431	0.296
22	Market Cap KES	23.24735	0.226
22	Zero Volume Share	21.61293	0.210
22	Sector Return Available	21.51049	0.209
22	Market Return MCW	21.07149	0.205
22	Zero Return Share	16.16169	0.157
22	HHI	5.36051	0.052
22	Sector Return EW	1.59181	0.015
22	Sector Return MCW	0.60686	0.006

Note. Gain measures the cumulative improvement in the XGBoost objective attributable to splits using each feature, after aggregation to the original feature name. Gain share reports that feature's percentage of total aggregated gain within the corresponding forecast horizon and sums to approximately 100% within each horizon. Raw gain values should therefore be interpreted within, rather than directly across, horizons. Feature importance is descriptive and does not establish causal effects.

Variable labels. EVT_u = EVT threshold; EVT_pu = exceedance probability at the threshold; EVT_nhist = historical EVT estimation count; EVT_nexc = number of exceedances; GARCH_Sigma = conditional volatility; GARCH_Mu = conditional mean; StdInnovation = standardised innovation.

The gain rankings change materially across forecast horizons. At one day, the largest shares are associated with the EVT threshold, exceedance probability, sector return, EVT history, standardised innovation, and

GJR-GARCH volatility variables. At five days, EVT history and tail-state variables remain prominent, alongside sector returns and GJR-GARCH conditional mean and volatility. At 10 days, sector returns, EVT tail information, and GJR-GARCH volatility rank highest, while at 22 days, EVT exceedance probability and threshold record the two largest gain shares, followed by GJR-GARCH volatility, conditional mean, EVT history, and standardised innovation.

The changing rankings show that XGBoost draws predictive information from different combinations of tail, volatility, return, liquidity, dispersion, and market-state variables as the forecast horizon changes. This pattern is consistent with the component-ablation results, which also show horizon-dependent contributions rather than a single predictor group uniformly improving forecasts. The gain rankings remain descriptive, however, and should not be interpreted as causal effects or as evidence that an individual high-ranked variable is indispensable to forecast performance.

As a final reproducibility check, the study satisfied all pre-specified pipeline audits before interpreting the robustness results. The five-seed neural phase completed all six architectures, preserved coherent multi-horizon probabilities, maintained magnitude-aware graph scaling, and retained both outcome classes at every Test horizon. Event-safe crash handling also remained intact throughout the neural workflow, preventing the forecasting pipeline from altering or propagating missing crash labels.

Paired-seed and moving-block-bootstrap comparisons showed modest gains from dynamic graph conditioning at selected horizons, particularly 5 and 10 days, relative to the no-graph architecture. However, the corresponding confidence intervals included zero, and the dynamic model did not consistently outperform the random or static graph controls.

The robustness phase completed all planned component-ablation and crash-definition fits without pipeline failure. Those ablations showed that volatility, breadth and dispersion, and market-state information contribute selectively across models and forecast horizons.

The study estimated fixed thresholds using training data only, and separate integrity checks reproduced the frozen 2.5% primary EVT crash labels without discrepancy. Finally, the average Test-sample Jaccard overlap of approximately 0.41 shows that EVT and fixed-threshold rules identify materially different crash-event sets. This limited overlap indicates that volatility conditioning changes which observations are classified as crashes rather than simply relabeling the same extreme-return days.

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